# Technical training.
## Product information.

# F23 Complete Vehicle


Edited for the U.S. market by:
**BMW Group University**
**Technical Training**
ST1502 2/1/2015

**BMW Service**

# General information

### Symbols used

The following symbol is used in this document to facilitate better comprehension or to draw attention to very important information:

⚠

Contains important safety information and information that needs to be observed strictly in order to guarantee the smooth operation of the system.

### Information status and national-market versions

BMW Group vehicles meet the requirements of the highest safety and quality standards. Changes in requirements for environmental protection, customer benefits and design render necessary continuous development of systems and components. Consequently, there may be discrepancies between the contents of this document and the vehicles available in the training course.

This document basically relates to the European version of left hand drive vehicles. Some operating elements or components are arranged differently in right-hand drive vehicles than shown in the graphics in this document. Further differences may arise as the result of the equipment specification in specific markets or countries.

### Additional sources of information

Further information on the individual topics can be found in the following:

- Owner's Handbook
- Integrated Service Technical Application.

Contact: conceptinfo@bmw.de



The information contained in this document forms an integral part of the technical training of the BMW Group and is intended for the trainer and participants in the seminar. Refer to the latest relevant information systems of the BMW Group for any changes/additions to the technical data.

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**BV-72/Technical Training**

# F23 Complete Vehicle
## Contents

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## Contents

# F23 Complete Vehicle

# 1. Introduction

From March 2015 the BMW 2-Series Convertible will be added to the BMW 2-Series range. Like its predecessor, the BMW 1-Series Convertible E88, the F23 is a four-seater convertible with a fully automatic soft top.


BMW 2-Series Convertible

## 1.1. Models

The following models of the F23 will be available for the market launch in March 2015:

| Model | Engine | Performance [kW (HP)] | Torque [Nm (lb-ft)] | Displacement [$cm^3$] |
|---|---|---|---|---|
| BMW 228i Convertible | N20B20O0 | 180 (241) | 350 (258) | 1997 |
| BMW 228i xDrive Convertible | N20B20O0 | 180 (241) | 350 (258) | 1997 |
| BMW M235i Convertible | N55B30O0 | 240 (322) | 450 (332) | 2979 |

# F23 Complete Vehicle
# 1. Introduction

## 1.2. Special features

The F23 has the following special features that distinguish it from its predecessor:

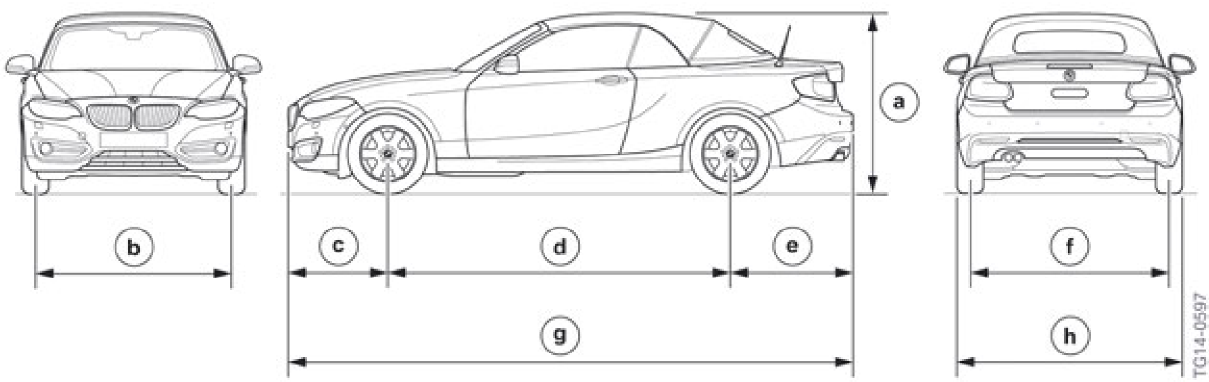
- Improved driving dynamics thanks to significantly increased rigidity of the front end
- Soft top with optimized sound insulation
- The soft top can be opened and closed while the car is being driven up to a speed of approximately 30 mph / 50 km/h
- Fully embedded rollover protection system
- Increased luggage compartment capacity
- Automatic engine start-stop function (MSA) on all models
- Electronic Power Steering (EPS) (electromechanical power steering) on all models
- Optional variable sport steering (Option 2VL)
- Reasonable axle-load distribution thanks to forward shift of the front axle by 10 mm while retaining the standardized engine position
- Increased front track width by 41 mm
- Optional M sports suspension (Option 704) or Adaptive M suspension (Option 2VF)
- Widened range of assistance systems

# F23 Complete Vehicle
# 1. Introduction

## 1.3. Dimensions

Compared with the predecessor model BMW F88 1-Series Convertible, the F23 BMW 2-Series Convertible has larger dimensions.

Set out below are the outer dimensions of the F23 with comparison values for the E88 BMW 128i.




F23 outer dimensions

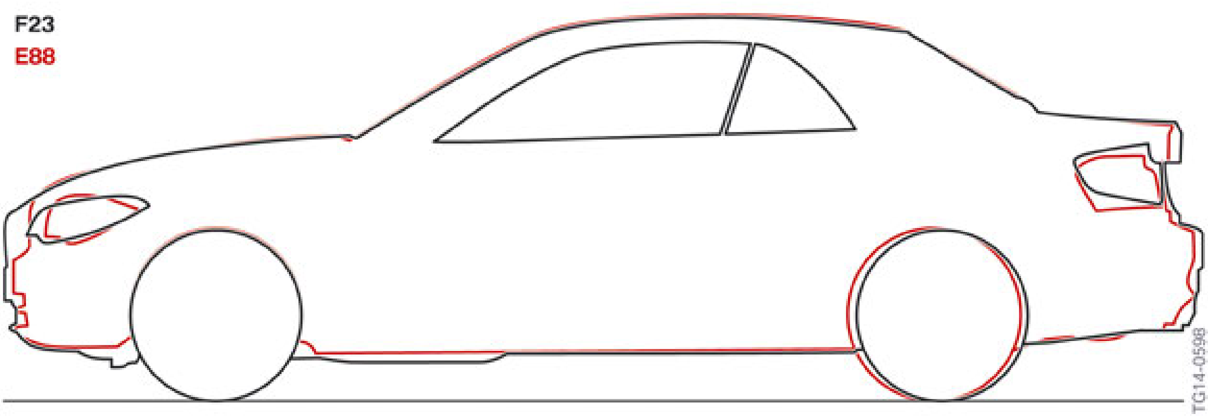
| Index | Explanation | | F23 BMW 228i Convertible | E88 BMW 128i Convertible |
|---|---|---|---|---|
| a | Vehicle height, empty | [mm] | 1413 | 1411 |
| b | Front track width, basic wheels | [mm] | 1521 | 1474 |
| c | Front overhang | [mm] | 785 | 751 |
| d | Wheelbase | [mm] | 2690 | 2660 |
| e | Rear overhang | [mm] | 962 | 949 |
| f | Rear track width, basic wheels | [mm] | 1556 | 1507 |
| g | Vehicle length | [mm] | 4437 | 4360 |
| h | Width excluding exterior mirrors (vehicle width incl. exterior mirrors) | [mm] | 1774 (1984) | 1748 (1919) |

### 1.4. Weight and payload

The vehicle curb weights and the payloads of the F23 are set out in the following table.

| Model | | Vehicle curb weight (US) Manual gearbox M235 only | Vehicle curb weight (US) Automatic transmission | Load capacity |
|---|---|---|---|---|
| BMW 228i Convertible | [lbs/ kg] | NA | 3625/1644 | 680/308 |
| BMW 228i xDrive Convertible | [lbs/ kg] | NA | 3765/1708 | 680/308 |
| BMW M235i Convertible | [lbs/ kg] | 3725/1690 | 3765/1708 | 680/308 |

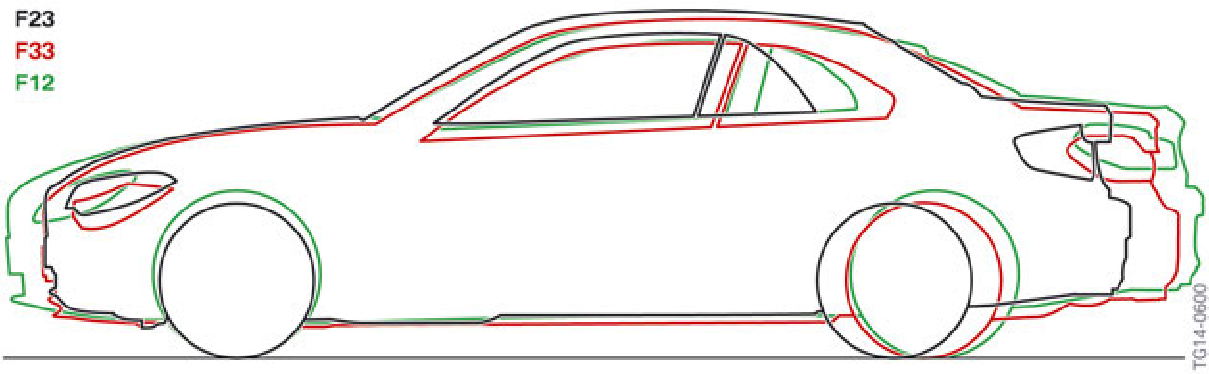
### 1.5. Silhouette comparison




F23 silhouette comparison with BMW 1-Series Convertible E88

| Explanation | | F23 BMW 228i Convertible | E88 BMW 128i Convertible |
|---|---|---|---|
| Vehicle height, empty | [mm] | 1413 | 1411 |
| Front overhang | [mm] | 785 | 751 |
| Wheelbase | [mm] | 2690 | 2660 |
| Rear overhang | [mm] | 962 | 949 |
| Vehicle length | [mm] | 4437 | 4360 |

# F23 Complete Vehicle
## 1. Introduction




F23 silhouette comparison with BMW 4-Series Convertible F33 and BMW 6-Series Convertible F12

| Explanation | | F23 BMW 228i Convertible | F33 BMW 428i Convertible | F12 BMW 640i Convertible |
|---|---|---|---|---|
| Vehicle height, empty | [mm] | 1413 | 1384 | 1365 |
| Front overhang | [mm] | 785 | 787 | 941 |
| Wheelbase | [mm] | 2690 | 2810 | 2855 |
| Rear overhang | [mm] | 962 | 1041 | 1098 |
| Vehicle length | [mm] | 4437 | 4638 | 4894 |

### 1.6. Material and color concept

In addition to the comprehensive offering of optional equipment, the F23 can also be individualized with the following equipment packages:

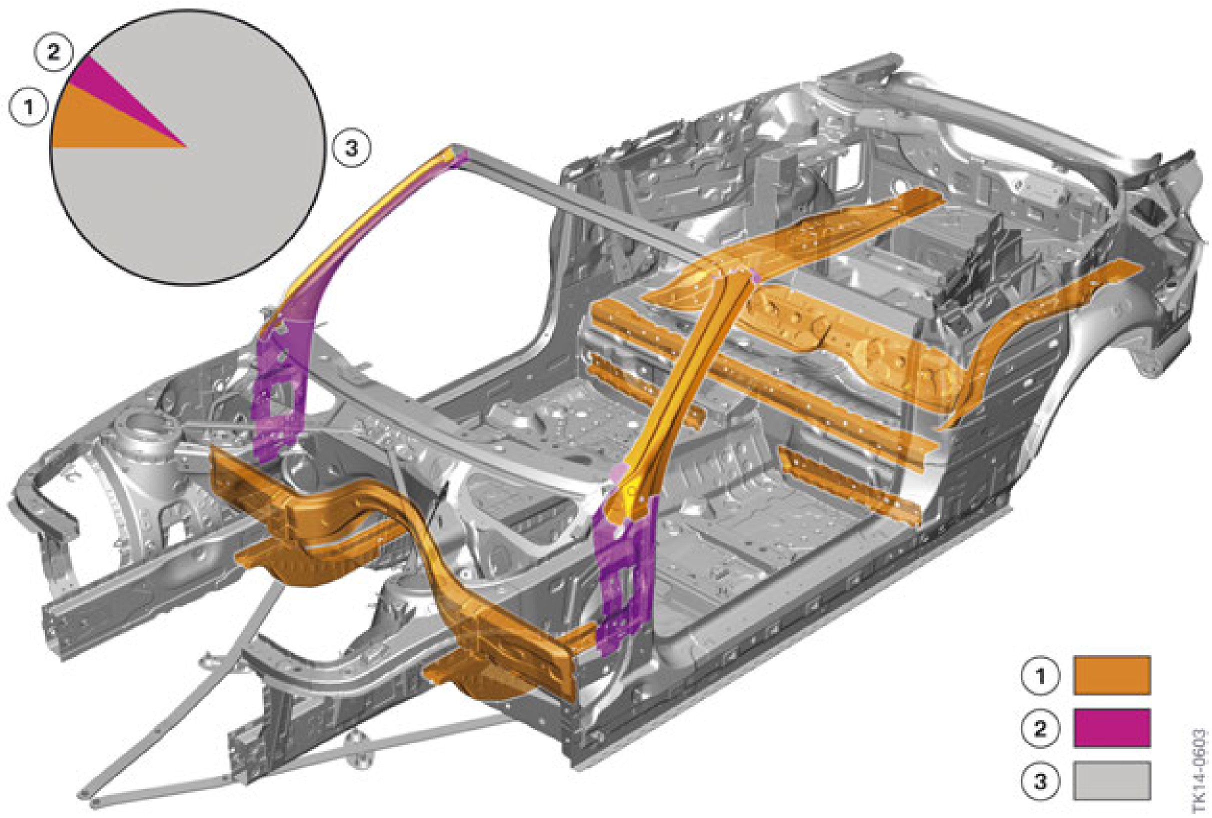
- M Sport Line (Option ZMM)
- Sport Line (Option ZSL)

# F23 Complete Vehicle

# 2. Body

## 2.1. Material properties of bodyshell

The use of state-of-the-art steels as well as the intelligent material mix have a positive effect on the vehicle weight. The distinctive body rigidity ensures the driving dynamics typical of BMW.




F23 material properties of bodyshell

| Index | Explanation |
|---|---|
| 1 | High-strength steels (> 300 MPa), 8 % of the body weight |
| 2 | Super-strength steels (> 900 MPa), 4 % of the body weight |
| 3 | Other steels (< 300 MPa), 88 % of the body weight |

In the case of high-strength steels the grain consist of several phases. They have a yield strength $Rp_{0.2}$ of 300 to 900 MPa. In the F23 these are for example:

- Dual-phase steels
- TRIP steels
- Complex-phase steels
- Martensitic-phase steels

Super-strength steels in the F23 are hot-formed manganese-boron steels with a yield strength $R_{P0.2}$ in excess of 900 MPa.

## 2.2. Interior equipment

### 2.2.1. Climate control

F23 control panel, IHKA 2/1 zones

The BMW 2-Series Convertible is fitted as standard with integrated automatic heating / air conditioning IHKA 2/1-Zone.

During a trip with the soft top open, a specific convertible program is activated in the heating and air conditioning system. A comfortable climate is guaranteed at very high or very low ambient temperatures.

Based on the good aerodynamics, comfortable driving is also possible at low ambient temperatures and a high driving speed in combination with the optional wind deflector.

Depending on the ambient temperature, driving speed and sunlight, the following are adapted by the convertible programme when the soft top is open:

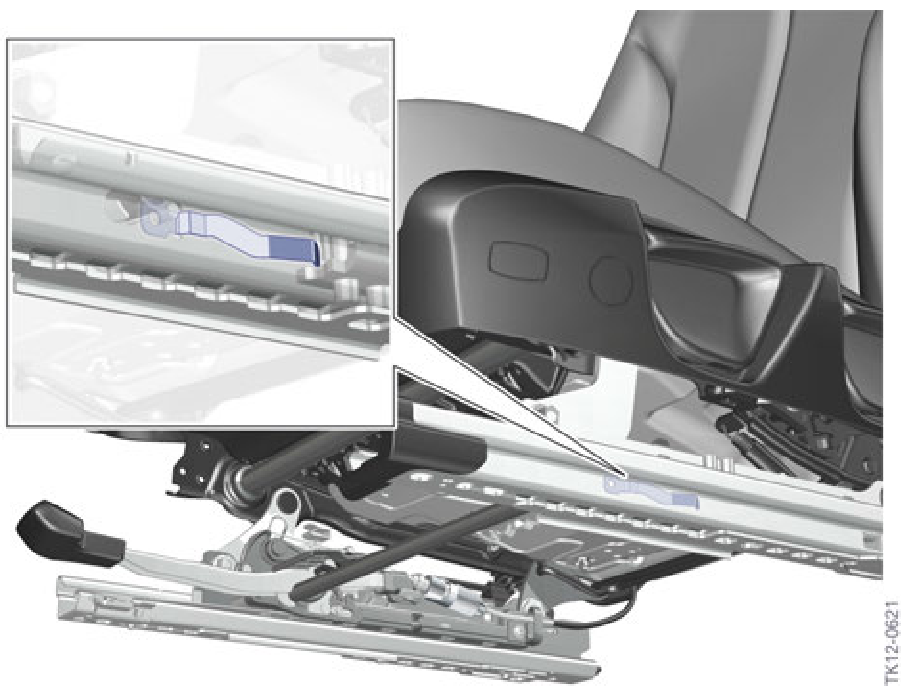
- air distribution and air temperature adapted.
- air recirculation function and automatic air recirculation control deactivated.
- values of condensation and interior temperature sensor are no longer considered.

### 2.2.2. Front seats

An integrated side airbag as well as a seat belt system with retractor tensioner is used for all front seat versions of the F23. The following optional equipment can be ordered:

- Seat adjustment, electrical, with memory function (Option 459, memory function only on driver's side) standard in the M325i.
- Sport seats for driver and front passenger (Option 481) standard in the M325i.
- Lumbar support for driver and front passenger (Option 488) standard in the M325i.
- Seat heating for driver and front passenger (Option 494).

**Rear easy-entry facility**




F23 seat rail adjustment (for manual seat adjustment)

In comparison to the predecessor E88 the spring which pulled the seat forwards for the rear easy-entry facility has been removed in the F23 with manual seat adjustment. Instead an adjustment which holds the seat in the forward position is now located in the seat rail.

# F23 Complete Vehicle
## 2. Body


F23 rear easy-entry facility (for electric seat adjustment, Option 459)

The additional one-touch function to change the entrance area in the F23 with electric seat adjustment with memory (Option 459) is also new:

**Comfort entrance:** Press the front arrow of the rocker switch briefly to move the front seat automatically to the end position. Press the button again stops the seat's movement.

**Original position:** Press the back arrow of the rocker switch briefly to move the front seat automatically to its original position. Press the button again stops the seat's movement.

### 2.2.3. Rear seats


F23 seat bench

The one-part seat bench in the BMW 2-Series Convertible has two contoured seats, between which there is a storage compartment. The rear seat backrest can be folded down to make use of the through-loading system.

## 2.3. Luggage compartment


F23 luggage compartment

Compared with the predecessor, in the F23 the load opening between the rear lights has been widened by 35 mm and the luggage compartment capacity has been increased:

- Open soft top: increase by approx. 20 liters to 280 liters in total.
- Closed soft top: increase by approx. 30 liters to 335 liters in total.

Compared with the E88, the opening of the through-loading system has been widened by 150 mm and is 28 mm taller.

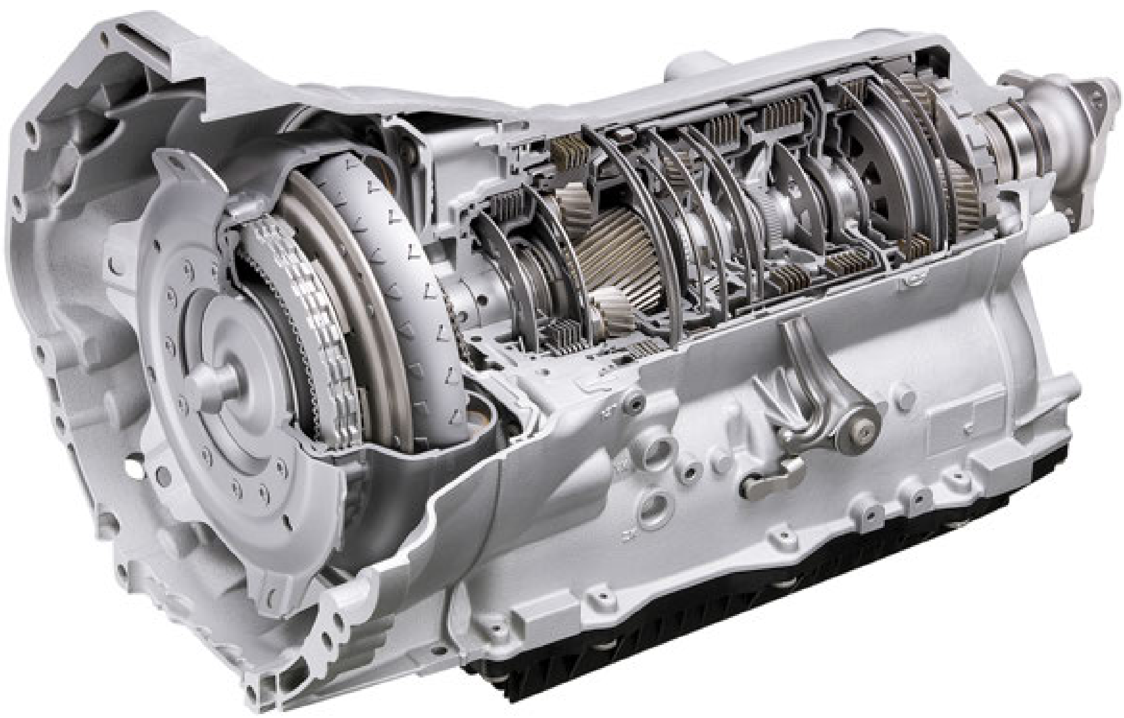
# F23 Complete Vehicle

# 3. Drivetrain

## 3.1. Automatic transmission

In the F23 uses the automatic transmission GA8HP45Z.

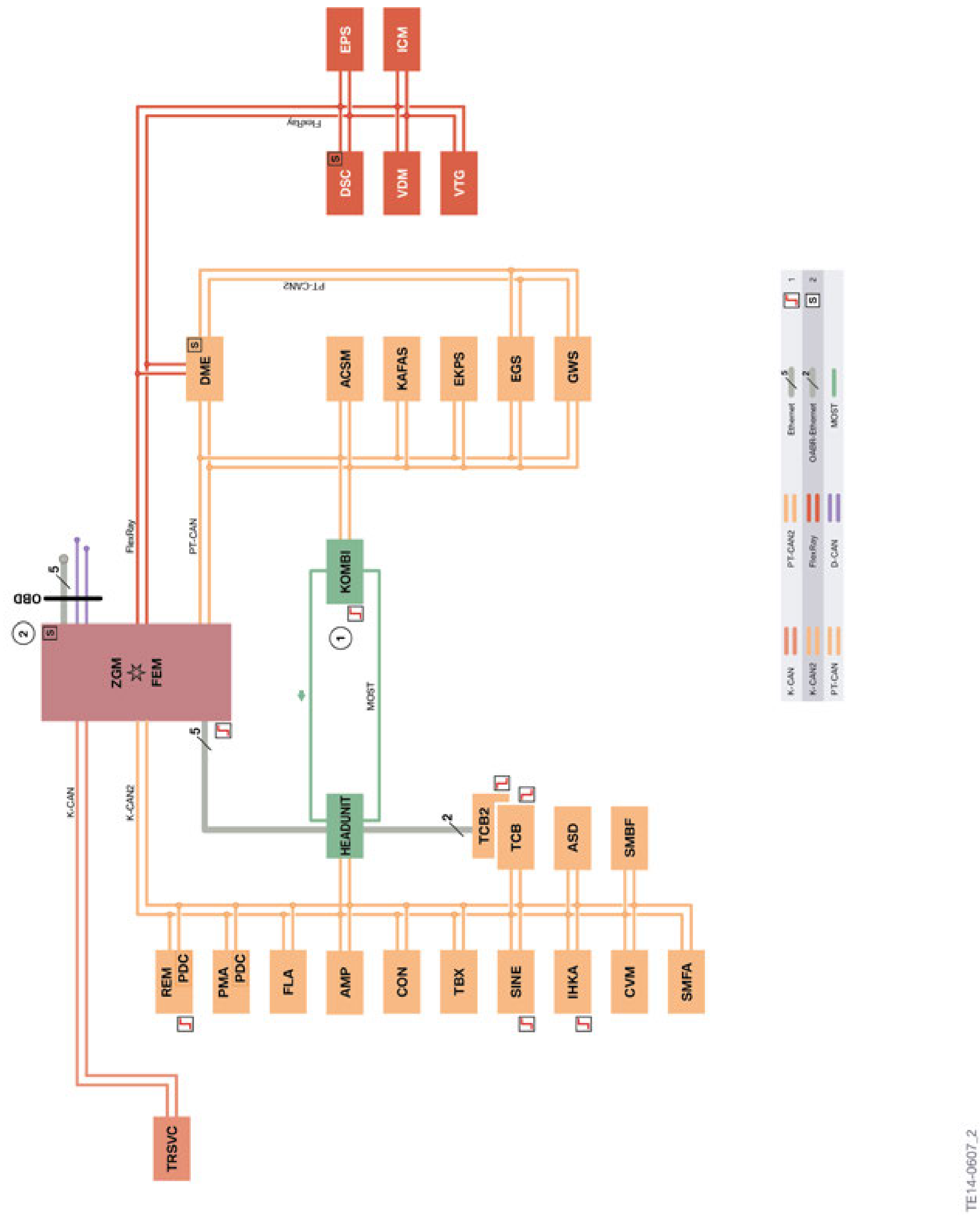
For more information on the transmission, please refer to Technical Training Reference Manual "ST1113 F30 Complete Vehicle".




GA8HP45Z Transmission

## 4. General Vehicle Electronics

### 4.1. Bus overview




F23 data bus overview

# F23 Complete Vehicle
## 4. General Vehicle Electronics

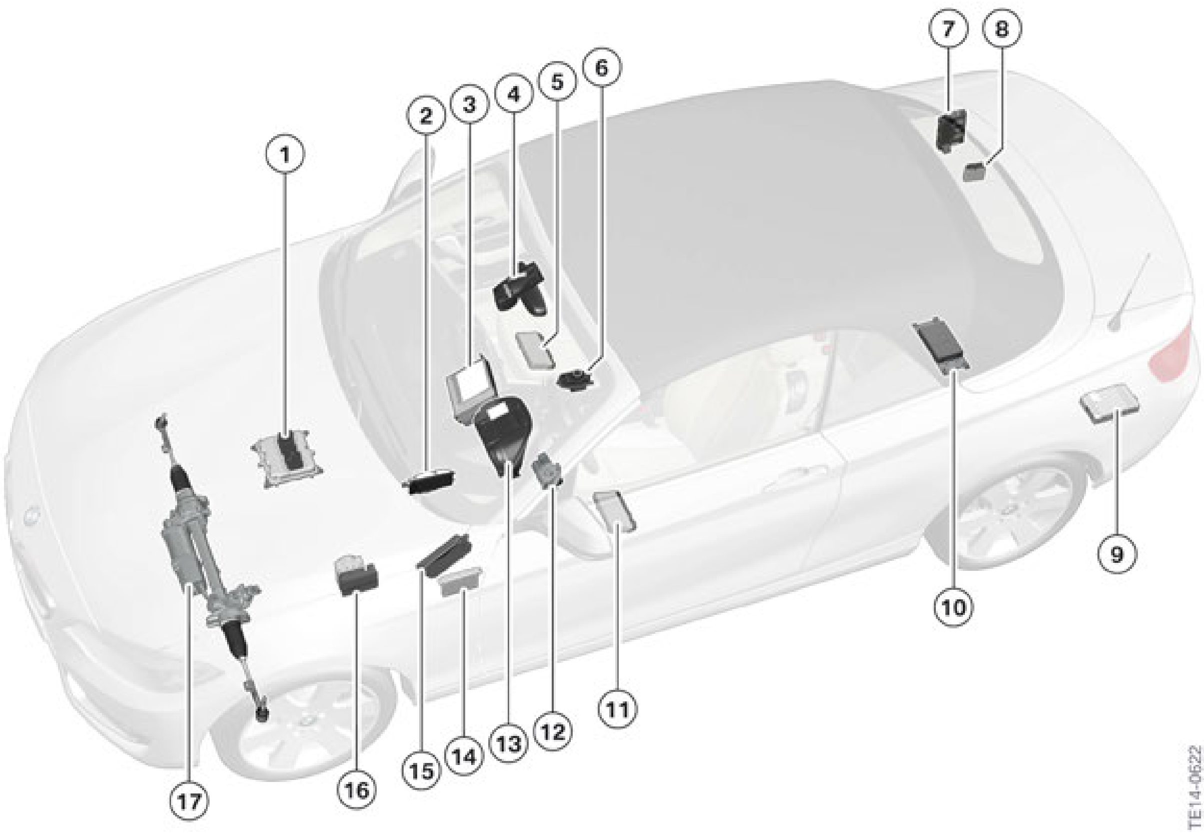
| Index | Explanation |
|---|---|
| 1 | Start-up nodes; control units for starting and synchronizing the FlexRay bus system |
| 2 | Control units with wake-up authorization |
| ACSM | Crash Safety Module |
| AMP | Amplifier |
| ASD | Active Sound Design |
| CON | Controller |
| CVM | Soft top module |
| DME | Digital Motor Electronics |
| DSC | Dynamic Stability Control |
| EGS | Electronic transmission control |
| EKPS | Electronic fuel pump control |
| EPS | Electronic Power Steering |
| FEM | Front Electronic Module |
| FLA | High-beam assistant |
| GWS | Gear selector |
| HEADUNIT | Headunit |
| ICM | Integrated Chassis Management |
| IHKA | Integrated automatic heating / air conditioning |
| KAFAS | Camera-based driver support systems |
| KOMBI | Instrument cluster |
| OBD | Diagnostic socket |
| PDC | Park Distance Control (with Option 5DP, Park Maneuvring Assistant integrated in the PMA control unit; otherwise integrated in the Rear Electronic Module) |
| PMA | Park Maneuvring Assistant |
| REM | Rear Electronic Module |
| SINE | Siren with tilt alarm sensor |
| SMBF | Seat module, passenger |
| SMFA | Seat module, driver |
| TBX | Touchbox |
| TCB | Telematic Communication Box |
| TCB2 | Telematic Communication Box 2 |

| Index | Explanation |
|---|---|
| TRSVC | Control unit for rear view camera and SideView |
| VDM | Vertical Dynamics Management |
| VTG | Transfer box |
| ZGM | Central gateway module |

### 4.2. Complete overview, control units

The installation locations of the control units are shown in the following graphics:

#### 4.2.1. Front view




F23 installation location of the control units

| Index | Explanation |
|---|---|
| 1 | Digital Motor Electronics (DME) |
| 2 | Crash Safety Module (ACSM) |
| 3 | Headunit |
| 4 | High-beam assistant (FLA) |

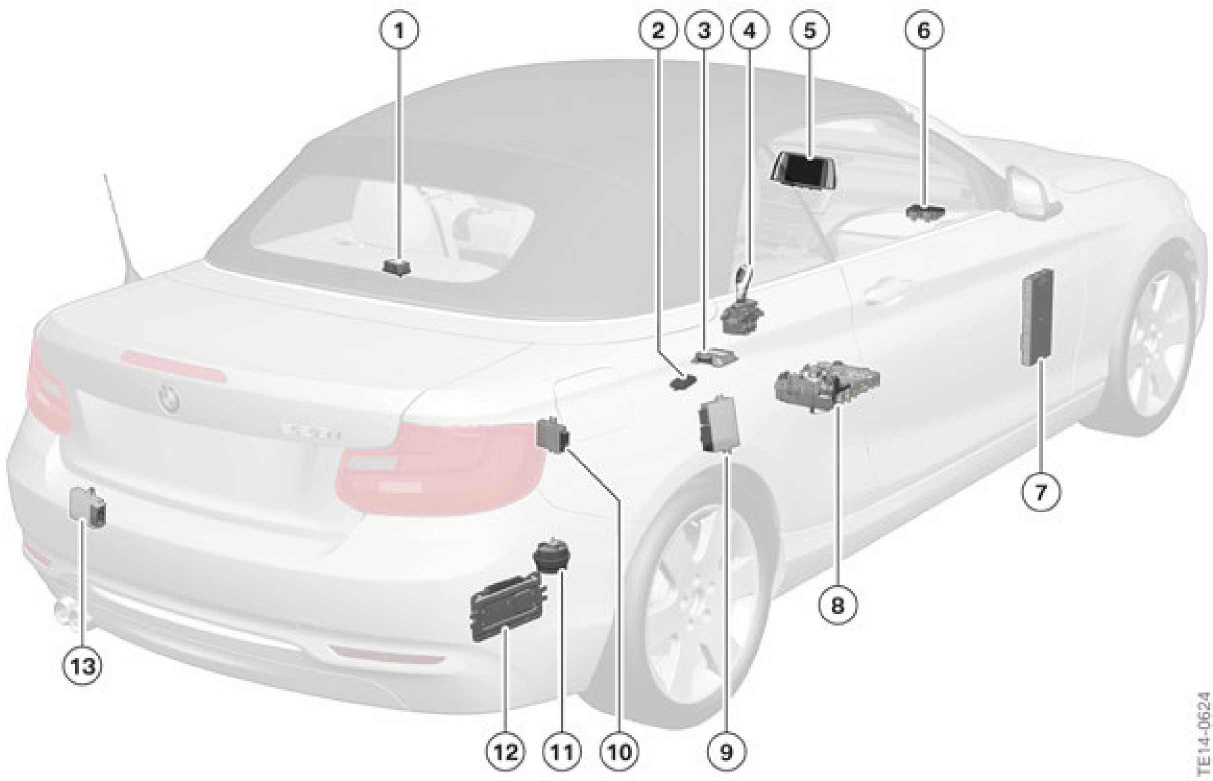
# 4. General Vehicle Electronics

| Index | Explanation |
| --- | --- |
| 5 | Front passenger seat module, (SMBF) |
| 6 | Controller (CON) |
| 7 | Rear Electronic Module (REM) |
| 8 | Park Maneuvring Assistant (PMA) |
| 9 | Amplifier (AMP) |
| 10 | Telematic Communication Box (TCB)/Telematic Communication Box 2 (TCB2) |
| 11 | Driver's seat module (SMFA) |
| 12 | Transfer box |
| 13 | Instrument cluster (KOMBI) |
| 14 | Camera-based driver support systems (KAFAS) |
| 15 | Control unit for rear view camera and SideView (TRSVC) |
| 16 | Dynamic Stability Control (DSC) |
| 17 | Electronic Power Steering (electromechanical power steering) (EPS) |

# F23 Complete Vehicle

## 4. General Vehicle Electronics

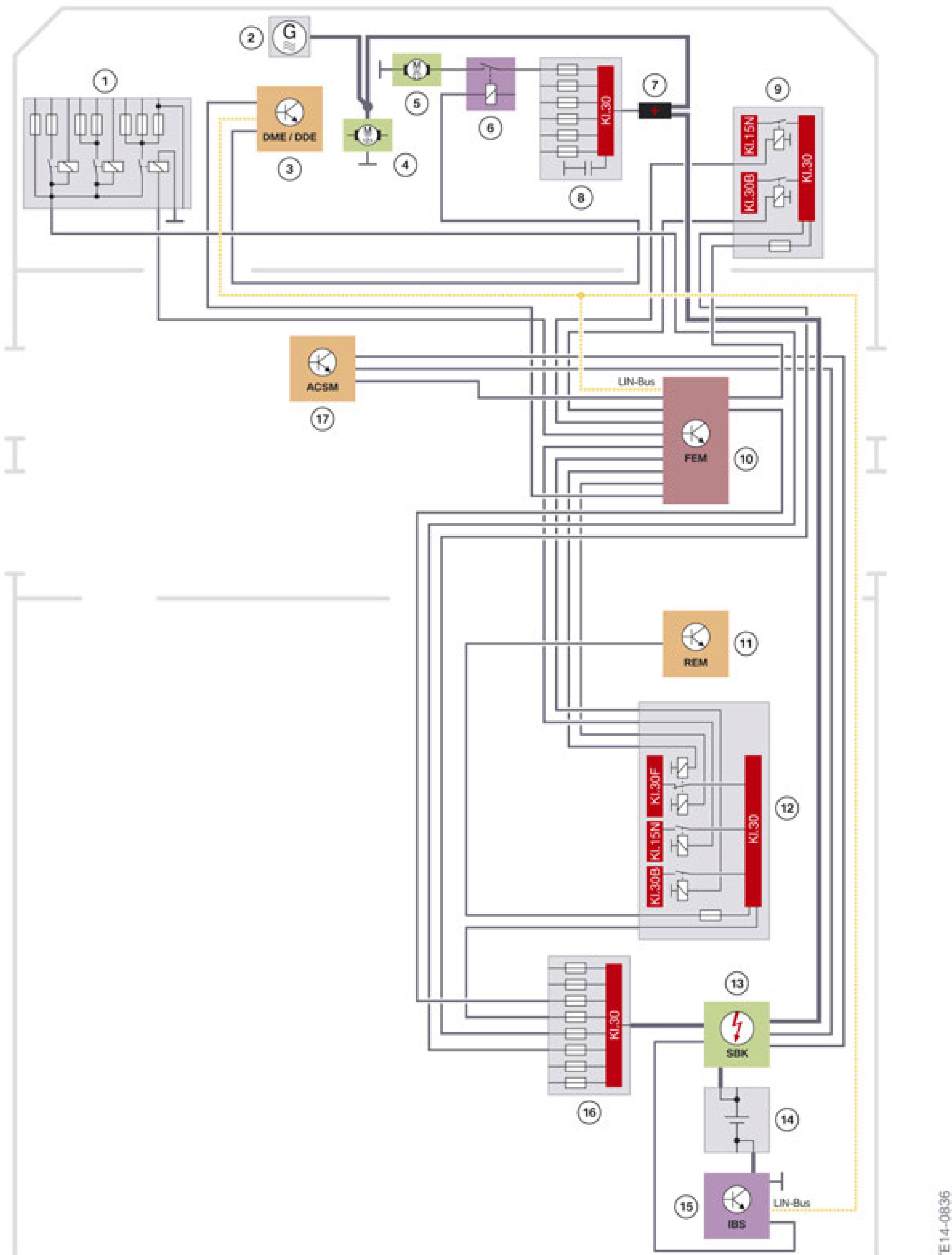
### 4.2.2. Rear View




F23 Installation location of the control units

| Index | Explanation |
|---|---|
| 1 | Remote control receiver |
| 2 | Touchbox (TBX) |
| 3 | Integrated Chassis Management (ICM) |
| 4 | Gear selector switch (GWS) |
| 5 | Central information display (CID) |
| 6 | Integrated automatic heating / air conditioning (IHKA) |
| 7 | Front Electronic Module (FEM) |
| 8 | Electronic transmission control (EGS) |
| 9 | Soft top module (CVM) |
| 10 | Electronic fuel pump control (EKPS) |
| 11 | Siren with tilt alarm sensor (SINE) |
| 12 | Vertical Dynamics Management (VDM) |
| 13 | Active Sound Design (ASD) |

### 4.3. Voltage supply




F23 Voltage supply

# 4. General Vehicle Electronics

| Index | Explanation |
|---|---|
| 1 | Power Distribution Module (PDM) |
| 2 | Alternator |
| 3 | Digital Motor Electronics (DME) |
| 4 | Starter motor |
| 5 | Electric fan |
| 6 | Relay for electric fan |
| 7 | Positive battery connection point |
| 8 | Power distribution box, front |
| 9 | Power distribution box, engine compartment |
| 10 | Front Electronic Module (FEM) |
| 11 | Rear Electronic Module (REM) |
| 12 | Power distribution box, luggage compartment |
| 13 | Safety battery terminal (SBK) |
| 14 | Battery |
| 15 | Intelligent battery sensor (IBS) |
| 16 | Battery power distribution box |
| 17 | Crash Safety Module (ACSM) |

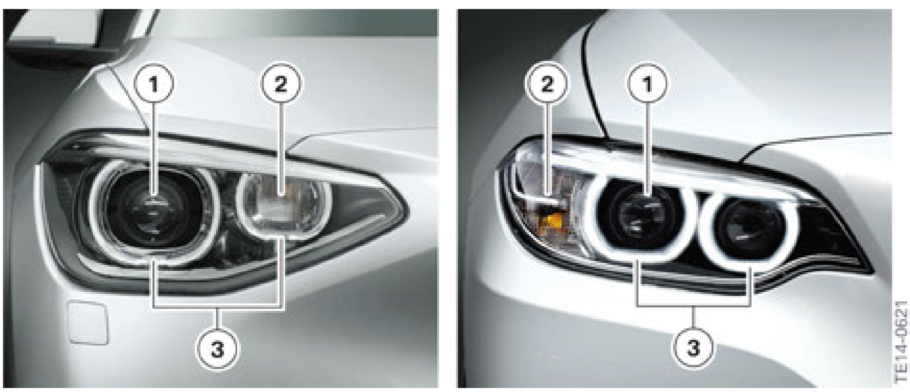
# F23 Complete Vehicle
# 4. General Vehicle Electronics

## 4.4. Headlights

The F23 228i and 228i xDrive is fitted with halogen headlights as standard. The following optional equipment is available for the F23 headlights:

- Xenon light (Option 522) standard in M235i.
- Adaptive Headlight (Option 524) standard in M235i.
- Fog light (not available with the M235i).
- High-beam assistant (Option 5AC).

The headlights of the F23 differ in shape from those of the F20 (Not for US). The turn indicator is now located in an outer compartment of the headlight.




Comparison of F20 (not US) xenon headlight with F23

| Index | Explanation |
|---|---|
| 1 | Bi-xenon headlight |
| 2 | Turn indicator |
| 3 | Side light/daytime driving light |

### 4.4.1. Halogen headlights


F23 halogen headlights

| Index | Explanation |
|---|---|
| 1 | High beam/daytime driving light |
| 2 | Side light |
| 3 | Low-beam headlight |
| 4 | Turn indicator |

### 4.4.2. Xenon headlight

The Adaptive Headlight (Option 524) can also be ordered in conjunction with the optional xenon headlights.

**The following functions are integrated in the optional equipment Adaptive Headlight (Option 524):**

- Adaptive headlight
- Cornering lights

**Cornering lights**

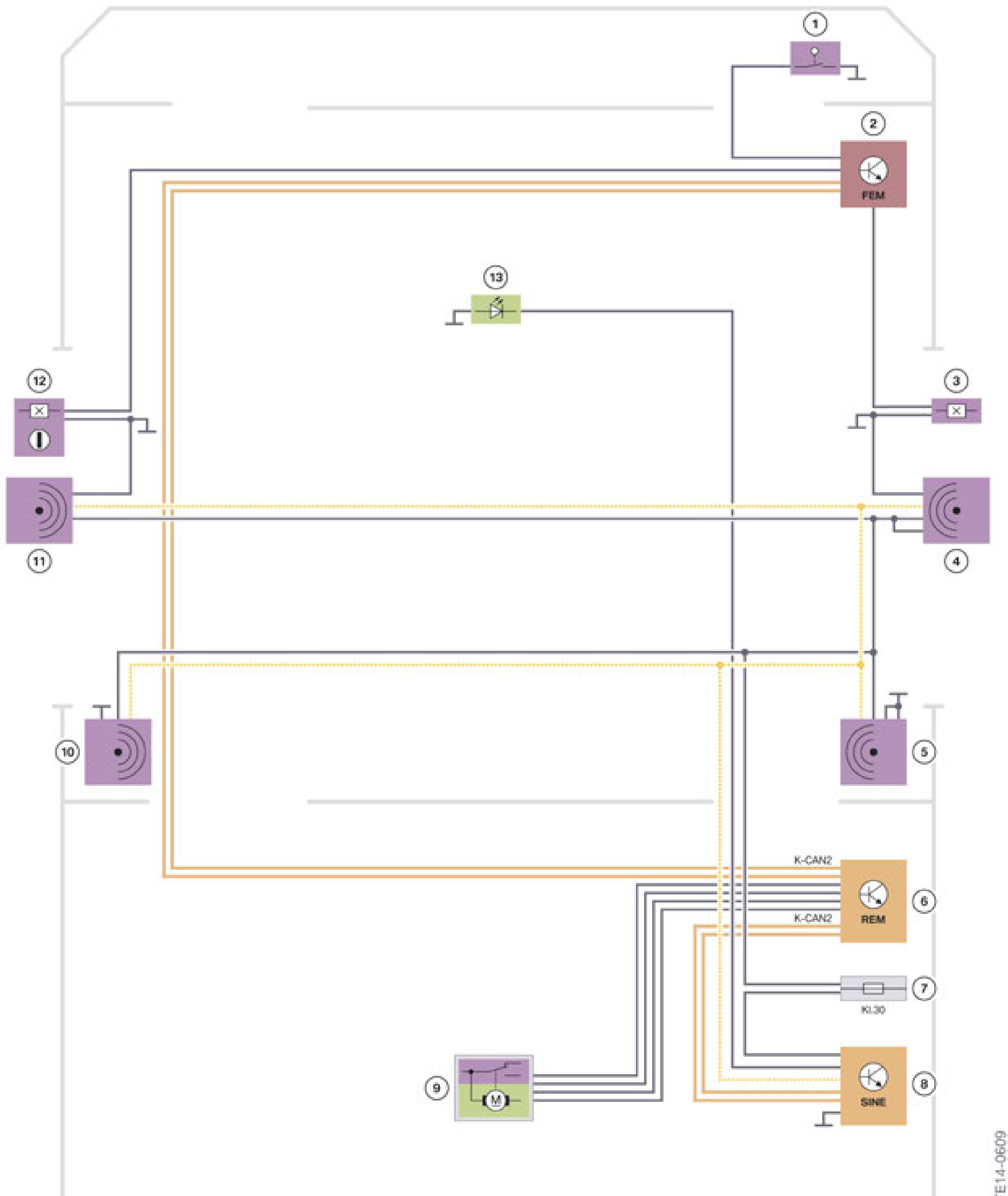
Cornering lights are only available in conjunction with xenon headlights with Adaptive Headlight (Option 524) and are integrated in the fog lights. Since the M235i does not have fog lights, the M235i will not have cornering lights.


F23 xenon headlight

| Index | Explanation |
|---|---|
| 1 | Fog light/cornering light in conjunction with optional equipment Adaptive Headlight (Option 524) |
| 2 | Side light/daytime driving light |
| 3 | Turn indicator |
| 4 | Bi-xenon headlight (low/high beam) |
| 5 | Accent light |

### 4.5. Alarm system (Option 302)

#### 4.5.1. System wiring diagram

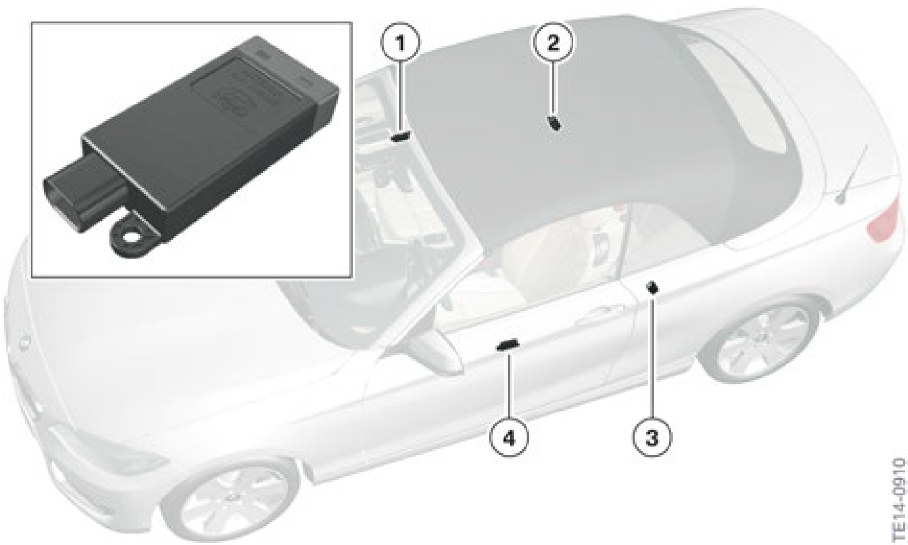



F23 system wiring diagram, alarm system

| Index | Explanation |
|---|---|
| 1 | Engine compartment lid contact switch |
| 2 | Front Electronic Module (FEM) |
| 3 | Door contact, front passenger side |
| 4 | Microwave sensor, passenger's door |
| 5 | Microwave sensor, rear right |
| 6 | Rear Electronic Module (REM) |
| 7 | Power distribution box, luggage compartment |
| 8 | Siren with tilt alarm sensor (SINE) |
| 9 | Trunk contact with trunk lock |
| 10 | Microwave sensor, rear left |
| 11 | Microwave sensor, driver's door |
| 12 | Door contact, driver's side |
| 13 | LED, alarm system |

### 4.5.2. Function

An alarm system is available as optional equipment (Option 302) for the F23. As in most BMW convertibles, the vehicle interior is monitored by microwave sensors and not by ultrasonic sensors. The reason for this is the susceptibility of ultrasonic sound to wind when the soft top is open. Ultrasonic sensor and microwave sensors detect movements inside the vehicle according to the same principle.




F23 microwave sensors, alarm system

| Index | Explanation |
|---|---|
| 1 | Sensor, door, right |
| 2 | Sensor, side panel, rear right |
| 3 | Sensor, side panel, rear left |
| 4 | Sensor, door, left |

# F23 Complete Vehicle
# 5. Driver Assistance Systems

## 5.1. Overview

The following assistance systems are available for the F23:

- Collision warning with pedestrian warning and city braking function
- Driver Assistance Plus
- Dynamic Cruise control
- High-beam assistant
- Speed limit info
- Park Distance Control
- Parking Manoeuvring Assistant
- Rear view camera

## 5.2. Further information

All the assistance systems of the F23 are very much familiar from other BMW models. These system can be found in various Technical Reference Manuals.

| **Designation** |
|---|
| Collision warning with pedestrian warning and city braking function (contained in Option 5AS, Driving Assistant) |
| Driver Assistance Plus (contained in Option 5AS, Driving Assistant) |
| Dynamic cruise control |
| High-beam assistant (Option 5AC) |
| Adaptive Headlights (Option 524) |
| Speed limit info (Option 8TH) |
| Rear view camera (Option 3AG) |

## 5.3. Driving Assistant

The following assistance systems are contained in the optional equipment Driving Assistant Plus (Option ZDB) and cannot be ordered individually for the customers:

- Lane departure warning (optical and haptic warning)
- Camera-based collision warning with pedestrian warning and city braking function
- Active driving assistant

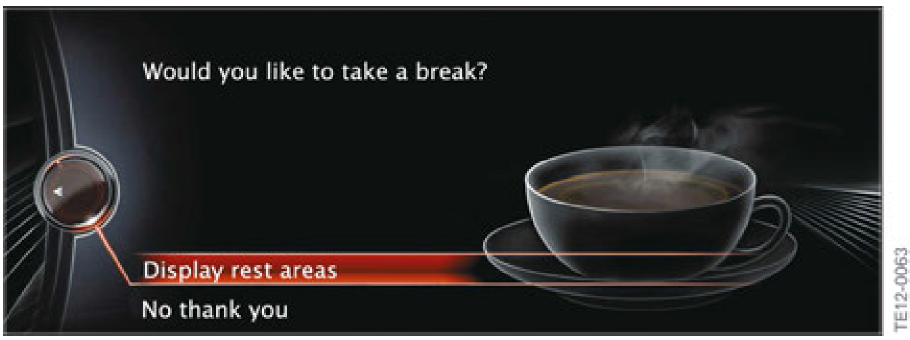
These assistance systems are operated by means of the buttons in the switch block.


F23 Intelligent Safety button

| Index | Explanation |
|---|---|
| 1 | Intelligent Safety button |
| 2 | Lane departure warning button |

### 5.4. Alertness assistant

The alertness assistant is contained in the optional equipment Driving Assistant (Option 5AS) and helps to prevent accidents caused by tiredness during long, monotonous journeys. The alertness assistant recognizes a change in the driver's driving behavior. In the event of increasing inattentiveness or if the driver is tired, the alertness assistant shows a display recommending that the driver take a break as a Check Control message in the central information display (CID).




Alertness assistant display in the CID

#### 5.4.1. Function

The alertness assistant is automatically active after each engine start from a speed of roughly 43 mph / 70 km/h. It cannot be switched off. When the alertness assistant is active, an alertness profile is created at the start of a trip. The following points are taken into consideration:

- Personal driving style
- Steering behavior
- Time display
- Trip duration
- Speed

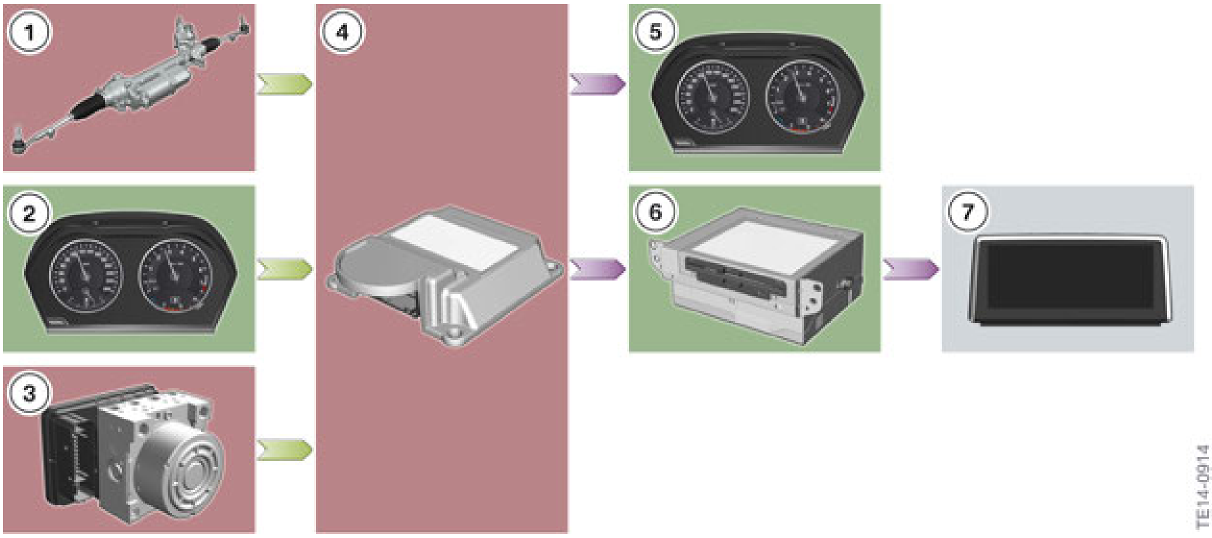
The driver's driving behavior changes if he is distracted or subject to increasing tiredness. If the changed driving behavior differs significantly from the taught-in driving behavior of the alertness profile, the alertness assistant shows a break recommendation in the central information display (CID).

This message must be acknowledged by the driver. In addition, he receives the option of having stop opportunities displayed.

The break recommendation is only displayed once during an uninterrupted journey. After a break a new break recommendation is not shown in the following 45 minutes of the trip. The system is reset if a break is longer than 50 minutes or if the break lasts longer than the last driving time. The driving behavior is then re-taught.

The alertness assistant is integrated in the Integrated Chassis Management (ICM).

# 5. Driver Assistance Systems




F23 Inputs/outputs, alertness assistant

| Index | Control unit | Function |
|---|---|---|
| 1 | Electronic Power Steering (EPS) | Steering behavior |
| 2 | Instrument cluster (KOMBI) | Time display |
| 3 | Dynamic Stability Control (DSC) | • Vehicle speed<br>• Yaw rate |
| 4 | Integrated Chassis Management (ICM) | Master control unit |
| 5 | Instrument cluster (KOMBI) | Display |
| 6 | Headunit | Indication in the CID via APIX interface |
| 7 | Central Information Display (CID) | Display |

### 5.4.2. Limits of the system

The alertness assistant cannot replace the personal assessment of the physical condition and may not be able to identify, or identify on time dwindling alertness or tiredness.

The function can be restricted for example in the following situations and an unwarranted break recommendation shown or no break recommendation shown at all:

- if the time is incorrectly set
- if the speed driven is mainly lower than 45 mph / 75 km/h
- in the event of a sporty driving style, e.g. heavy acceleration or quick cornering
- in active driving situations, e.g. frequent lane changes
- poor road conditions
- strong crosswind.

A change of driver is not detected by the system.

## 5.5. Park Distance Control (PDC)

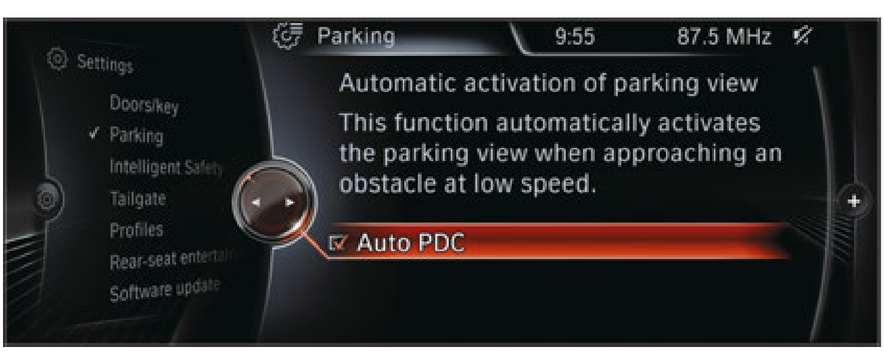
Visual and acoustic warning of obstacles is optimized in the F23. Park Distance Control cuts in with the Auto PDC function automatically; even when the vehicle is driving forwards.

### 5.5.1. Auto PDC

PDC is automatically activated with the Auto PDC function:

- if there is an object closer than 0.6 m to the front of the vehicle when driving forwards
- if there is an object closer than 1.5 m to the rear of the vehicle when reversing
- the speed driven does not exceed 2 mph / 4 km/h.

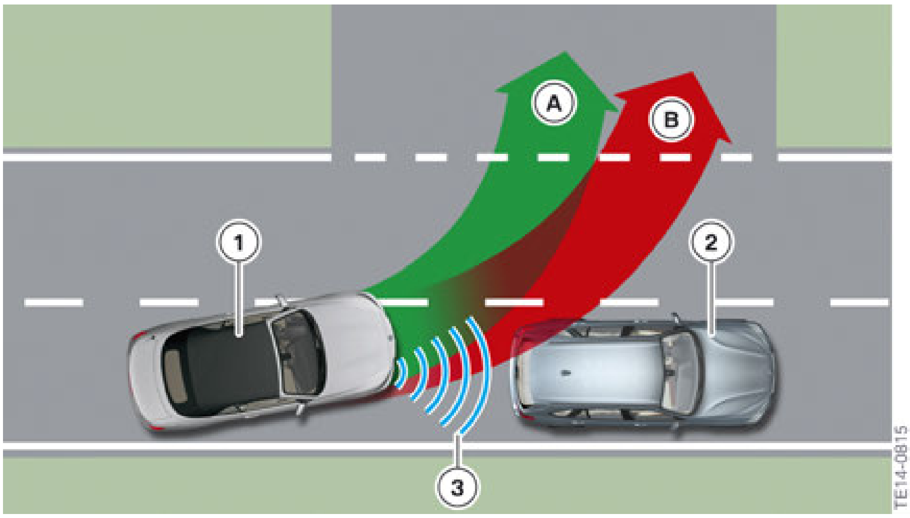
The automatic switching-on upon the identification of obstacles can be switched on and off in the "Settings" menu using a controller. The settings are stored for the ID transmitter currently used.




TE13-0866

F23 Auto PDC in the CID

A sound is only output if the detected object (in front of or behind the vehicle) is directly in the lane (risk of collision). If the detected object is not directly in the lane, the acknowledgement is only effected visually in the CID.




F23 object recognition

| Index | Explanation |
|---|---|
| A | Lane without object contact (**visual** acknowledgement in the CID) |
| B | Lane with object contact (**visual and acoustic** acknowledgement) |
| 1 | Turning vehicle |
| 2 | Detected object |
| 3 | Detection of obstacle by the PDC sensors |

If an object was detected by Auto PDC, the system must be reactivated so that it cannot be accidentally switched on for example in a car wash. Reactivation is effected by:

- Driving speed > 3 mph / 5 km/h
- Gear change or drive position change.

If a drive position change is made in vehicles with automatic transmission to the Neutral position, the system is not activated so that here too it cannot be accidentally switched on for example in a car wash.

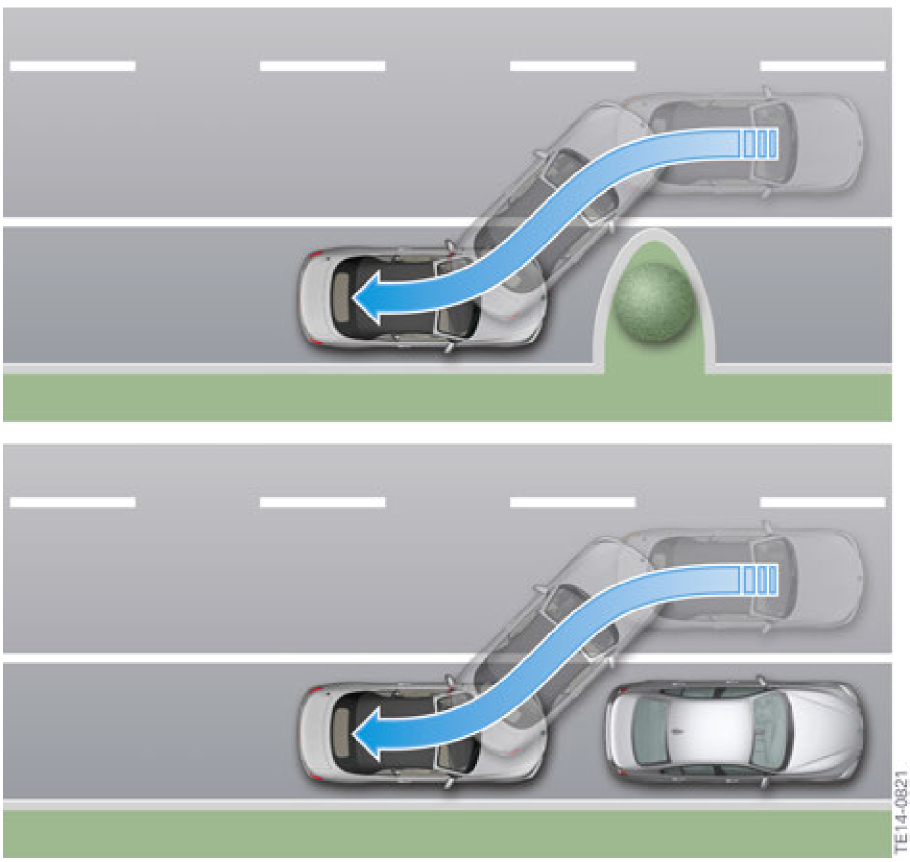
Auto PDC will be available shortly after the market launch for the F23.

## 5.6. Parking Maneuvring Assistant (PMA)

Whereas the first generation of the Parking Maneuvring Assistant (PMA) (Option 5DP) could only park parallel (to the roadway between two vehicles, the second generation of PMA can also park perpendicularly (cross parking) to the roadway.

Furthermore, it is no longer necessary for the parking space to be confined by two vehicles. With the second generation it is also possible to park behind only one object.

### 5.6.1. Curbside parking




F23 principle of curbside parking

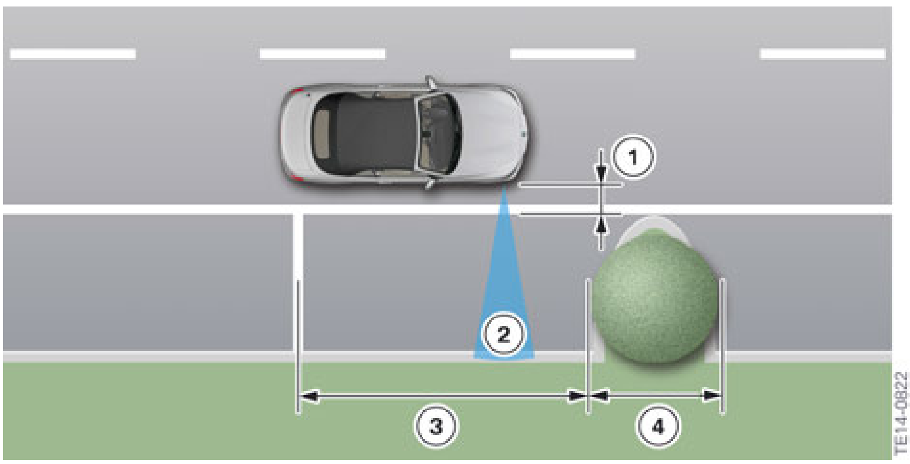
The following table shows the changes in parking space search between first and second generation PMA:

| Requirements | Parking Maneuvring Assistant before 03/15 | Parking Maneuvring Assistant after 03/15 |
|---|---|---|
| Size of parking space | Vehicle length + approx. 1.2 m | Vehicle length + approx. 0.8 m |
| Length of objects | min. 1.5 m | min. 0.5 m |
| Minimum number of detected objects | 2 | 1 |

# F23 Complete Vehicle
## 5. Driver Assistance Systems

The minimum length of an object detected by the vehicle no longer has to be 1.5 m, but instead only 0.5 m. With an object length of 0.5 m, a curb must still be detected by the PMA sensors for the purpose of parking in a parallel (curbside) parking space so as to ensure that the parking space is a parallel one.

The Parking Maneuvring Assistant of the F23 assumes solely the lateral guidance (steering operation), i.e. the driver himself has to brake and accelerate.

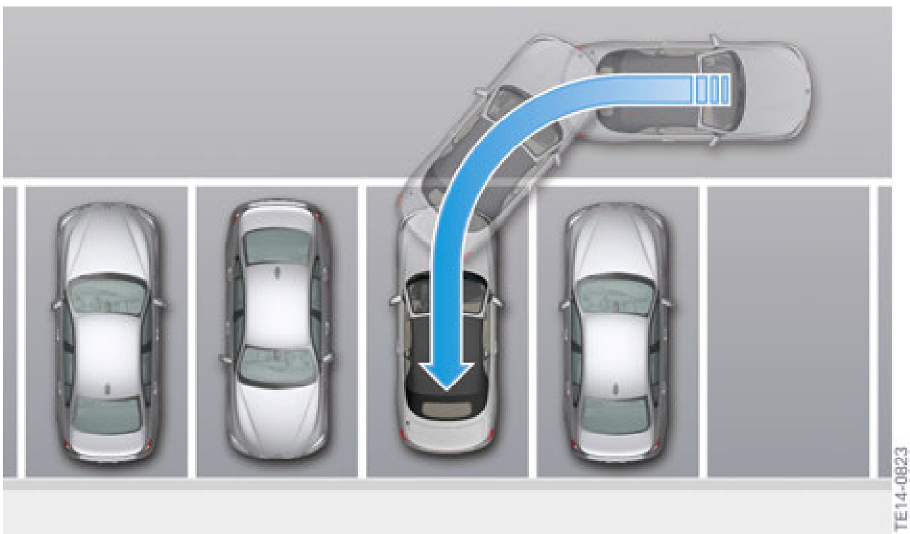



F23 requirement, parking spaces, curbside parking

| Index | Explanation |
|---|---|
| 1 | Maximum distance to the row of parked vehicles or objects: 1.5 m |
| 2 | Opening angle of the ultrasonic sensor horizontal: ± 40°, range approx. 4.2 m |
| 3 | Length of the parking space: vehicle length plus approx. 0.8 m |
| 4 | Object or vehicle length at least 0.5 m |

### 5.6.2. Cross parking

The cross parking function will be available for the F23 shortly after the market launch. Until this time the Parking Manoeuvring Assistant will only be offered with the curbside parking function.

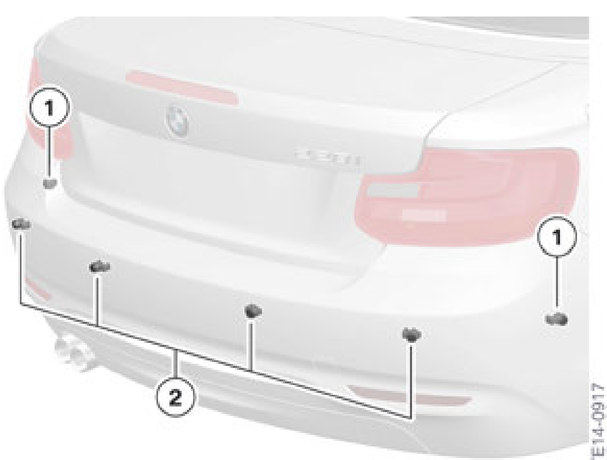



F23 principle of cross parking

The Parking Manoeuvring Assistant of the F23 assumes solely the lateral guidance (steering operation), i.e. the driver himself has to brake and accelerate.

#### Sensors

In the F23 two further ultrasonic sensors are installed in the rear bumper to detect precisely a perpendicular (cross) parking space during the parking Maneuvring. The ultrasonic sensors in the sides of the bumper on the F23 are PDC sensors, since they do not have to detect the depth of parking spaces, but instead only have to detect the distance to detected objects during the parking Manoeuvring.




F23 ultrasonic sensors in the rear bumper

| Index | Explanation |
|---|---|
| 1 | Ultrasonic sensors in rear bumper, side (PDC sensors) |
| 2 | Ultrasonic sensors in rear bumper (PDC sensors) |

### Parking space requirement

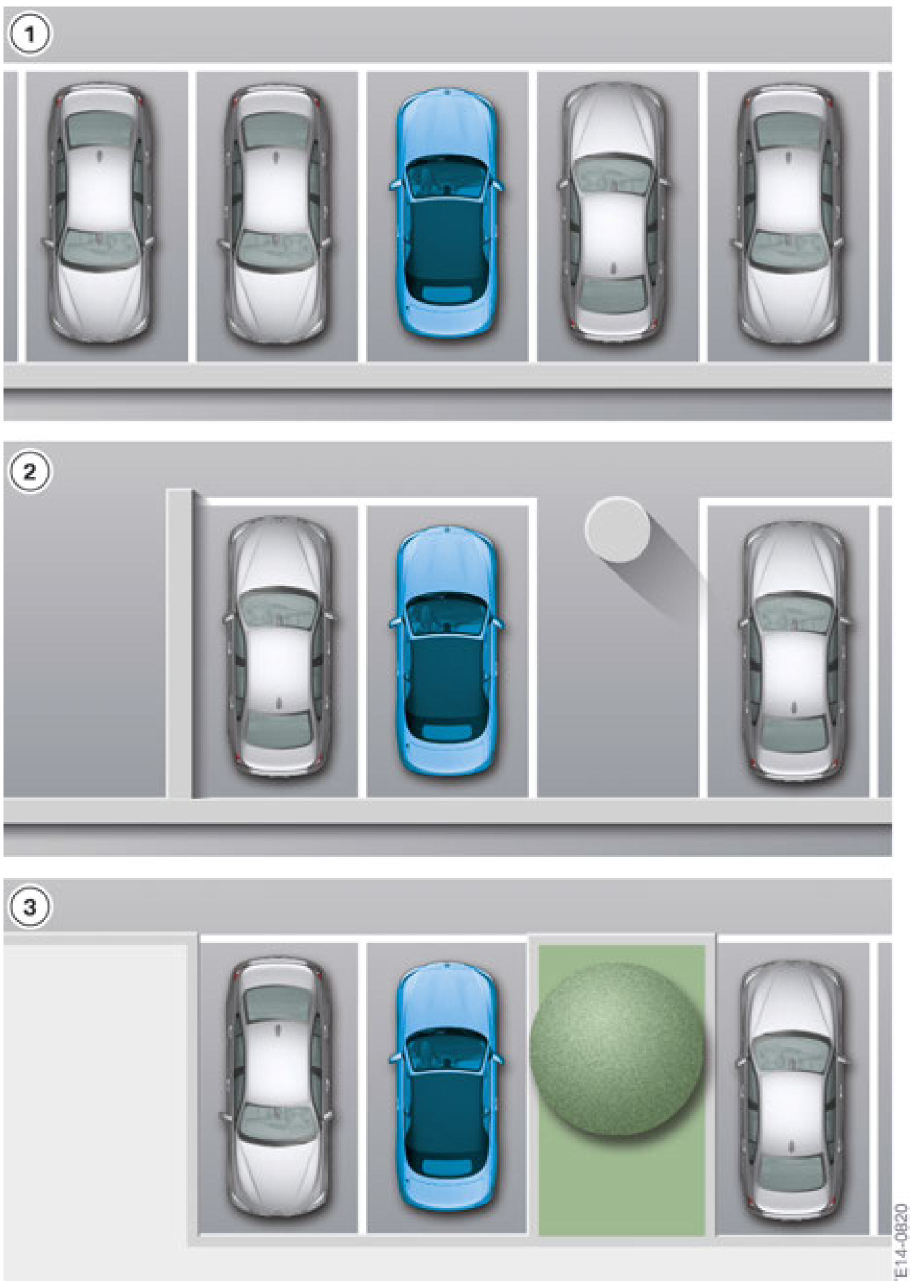

F23 requirement, parking space, cross parking

| Index | Explanation |
|---|---|
| 1 | Maximum distance to the row of parked vehicles: 1.5 m |
| 2 | Opening angle of the ultrasonic sensor horizontal: ± 40°, range approx. 4.2 m |
| 3 | Minimum depth of the parking space: own vehicle length |
| 4 | Vehicle or object width at least 0.5 m |
| 5 | Width of the parking space: vehicle width plus approx. 0.7 m up to max. 5 m |

### Different parking spaces

There are many different parking spaces that are perpendicular to the roadway. The F23 detects many of these parking spaces. Due to technical system limits the F23 cannot detect all the parking spaces.

The graphic below shows some examples of the most frequently arising parking spaces.




F23 perpendicular (cross) parking spaces

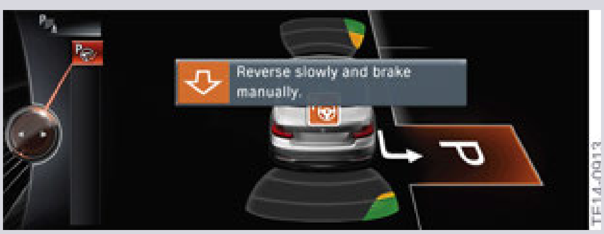
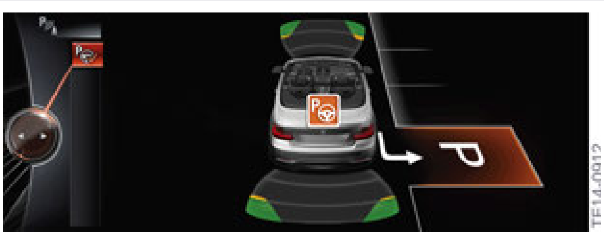
| Index | Explanation |
|---|---|
| 1 | Standard parking space between two cars |
| 2 | Parking space with post (underground car park) |
| 3 | Parking space with obstacles (public car park) |

⚠

The depth of perpendicular (cross) parking spaces must be gauged by the driver himself. Due to technical limits the system can only approximately determine the depth of perpendicular (cross) parking spaces.

### 5.6.3. Displays

Only the displays in the central information display (CID) during a cross parking manoeuvre are described below. The instructions in the CID for a parking Manoeuvring parallel to the roadway (curbside parking) are unchanged.

| Displays | Explanation |
|---|---|
|  | The Parking Maneuvring Assistant scans the surrounding area for a suitable parking space. |
|  | The driver should engage the reverse gear and take his hands off the steering wheel to start the parking process. |
|  | The driver is responsible for longitudinal guidance (acceleration and deceleration) of the vehicle. |
|  | The PMA is active and steers. |
|  | The parking Maneuvring is completed. The driver must secure the vehicle against rolling away. |

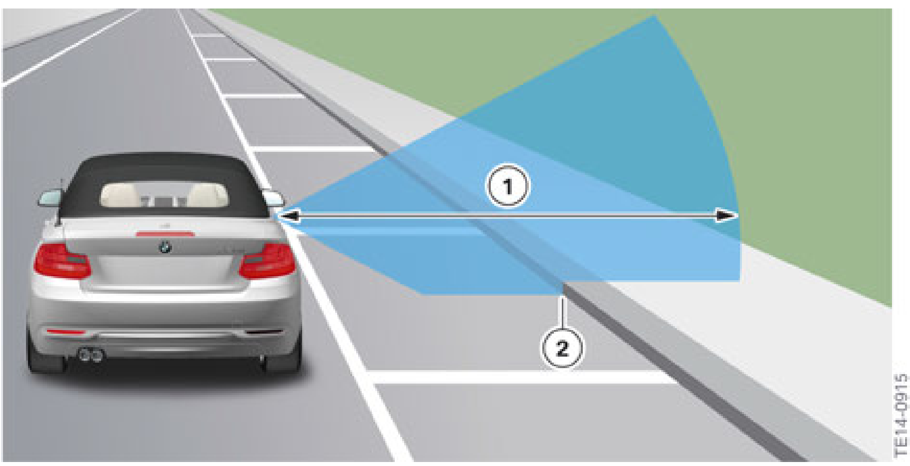
### 5.6.4. Universal parking space

Due to technical system limits the vehicle does not always detect the correct parking space. Situations may occur where parking spaces are wrongly detected or parking spaces that are not actually parking spaces are detected.

#### Curb detection

If a Curb is detected within the range of the ultrasonic sensor while the system is searching for a parking space, the parking space is for the most a space that is parallel to the road.

In the case of parking spaces that are perpendicular to the roadway, the curb is usually outside the detection range of the ultrasonic sensors (range approx. 4.2 m).



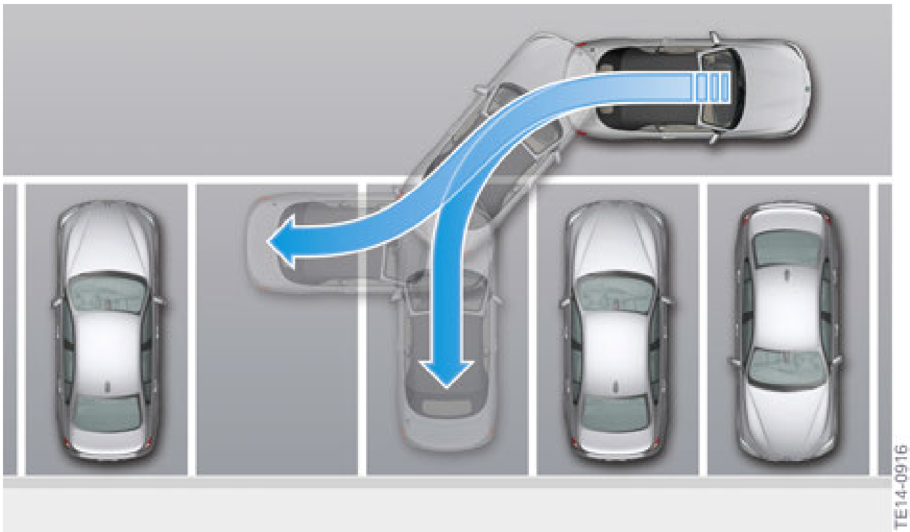
F23 curb detection by the ultrasonic sensors

| Index | Explanation |
|---|---|
| 1 | Range approx. 4.2 m; opening angle of ultrasonic sensors vertical ± 30° |
| 2 | Curb detection |

#### Detection of a universal parking space

The graphic below shows the detection of a universal parking space.

The detected parking space is of a size into which it would be possible to park both perpendicularly and parallel to the roadway. Because of the **absence** of a curb, the vehicle cannot correctly determine the parking space. The motor vehicle offers a universal parking space. The driver must decide how he wants to Manoeuvring into the parking space.




F23 detection of a universal parking space

### Display

The following display appears in the CID when the vehicle has detected a universal parking space. The driver can choose one parking option.

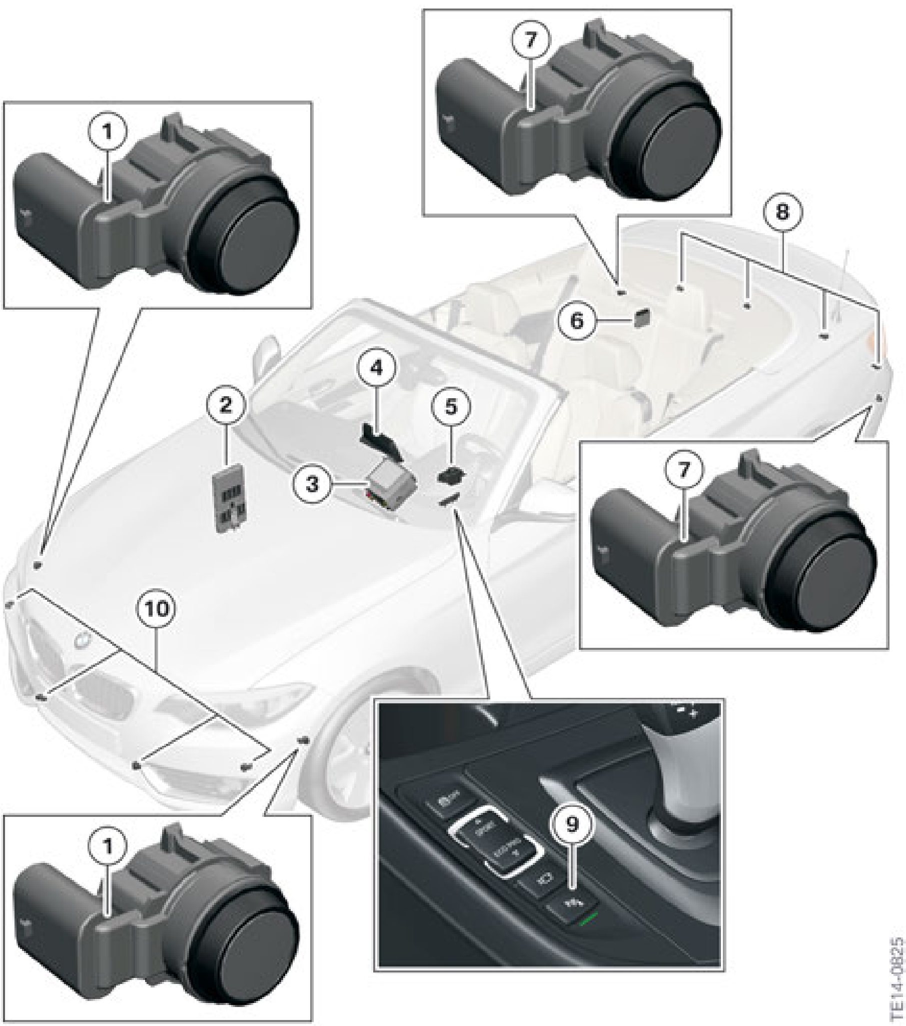

F23 universal parking space

| Index | Explanation |
|---|---|
| 1 | Perpendicular (cross) parking space |
| 2 | Parallel (curbside) parking space |

The table below shows the conditions under which the vehicle detects certain parking spaces:

| Detected object length | with detected curb | without detected curb |
|---|---|---|
| > 0.5 m (short) | Curbside parking | Cross parking |
| > 1.5 m (long) | Curbside parking | Universal parking space |

### 5.6.5. System components

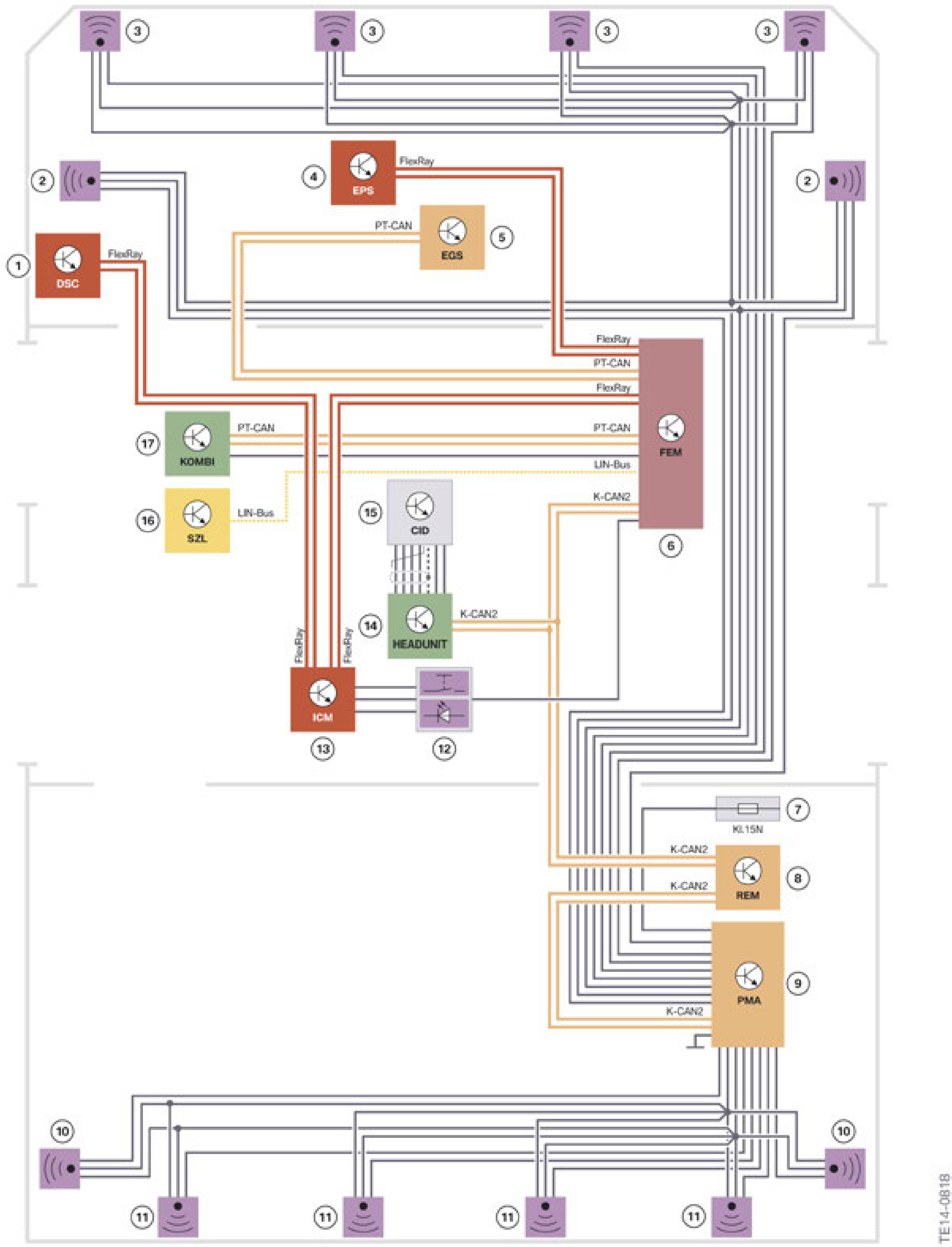



F23 System components of Parking Maneuvring Assistant

| Index | Explanation |
|---|---|
| 1 | Ultrasonic sensors of Parking Manoeuvring Assistant |
| 2 | Front Electronic Module (FEM) |
| 3 | Headunit |
| 4 | Central Information Display (CID) |

| Index | Explanation |
|---|---|
| 5 | Controller (CON) |
| 6 | Parking Manoeuvring Assistant (PMA) |
| 7 | Ultrasonic sensors, Park Distance Control, rear side |
| 8 | Ultrasonic sensors, Park Distance Control, rear |
| 9 | Park Distance Control button |
| 10 | Ultrasonic sensors for Park Distance Control, front |

### 5.6.6. System wiring diagram




F23 system wiring diagram for Parking Maneuvring Assistant

# 5. Driver Assistance Systems

| Index | Explanation |
|---|---|
| 1 | Dynamic Stability Control (DSC) |
| 2 | Ultrasonic sensors of Parking Manoeuvring Assistant |
| 3 | Ultrasonic sensors for Park Distance Control |
| 4 | Electronic Power Steering (electromechanical power steering) (EPS) |
| 5 | Electronic transmission control (EGS) |
| 6 | Front Electronic Module (FEM) |
| 7 | Power distribution box, luggage compartment |
| 8 | Rear Electronic Module (REM) |
| 9 | Parking Manoeuvring Assistant (PMA) |
| 10 | Ultrasonic sensors for Park Distance Control |
| 11 | Ultrasonic sensors for Park Distance Control |
| 12 | Park Distance Control button |
| 13 | Integrated Chassis Management (ICM) |
| 14 | Headunit |
| 15 | Central Information Display (CID) |
| 16 | Steering column switch cluster (SZL) |
| 17 | Instrument cluster (KOMBI) |

# 6. Info and Communication Systems

## 6.1. Headunits

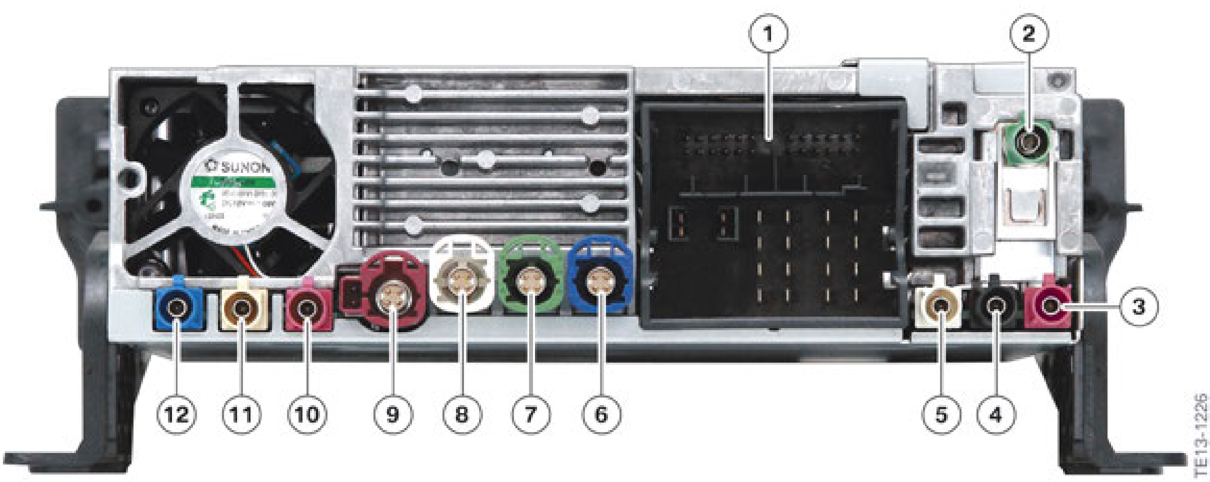
The F23 already features as standard the BMW Professional radio with a 6.5” central information display (CID).

### 6.1.1. Overview of headunits

| Equipment | Headunit | CID | Controller | Navigation |
|---|---|---|---|---|
| Radio BMW Professional (standard equipment) | Headunit Basis | 6.5” | 5-button | No |
| Navigation system Professional (Option 609) | Headunit High 2 | 8.8” | 7-buttons with touch control panel | Yes |

### 6.1.2. Headunit Basis

The Headunit Basis is installed in the F23 as standard equipment, which contains the BMW Professional radio (without navigation).




Rear view, Headunit Basis

| Index | Explanation |
|---|---|
| 1 | Connection, main connector |
| 2 | DAB L Band antenna (Not US) |
| 3 | SADARS satellite radio color code burgundy |
| 4 | FM1 antenna |
| 5 | FM2 antenna |
| 6 | USB2 connection; customer's smartphone via the telephone base plate |
| 7 | USB3 connection; connection to the Telematic Communication Box (TCB) |
| 8 | USB1 connection; customer access to USB audio interface |

| Index | Explanation |
|---|---|
| 9 | APIX connection to central information display (CID) |
| 10 | Preparation of WLAN antenna |
| 11 | Connection for Bluetooth antenna |
| 12 | GPS antenna |

### 6.1.3. Headunit High 2

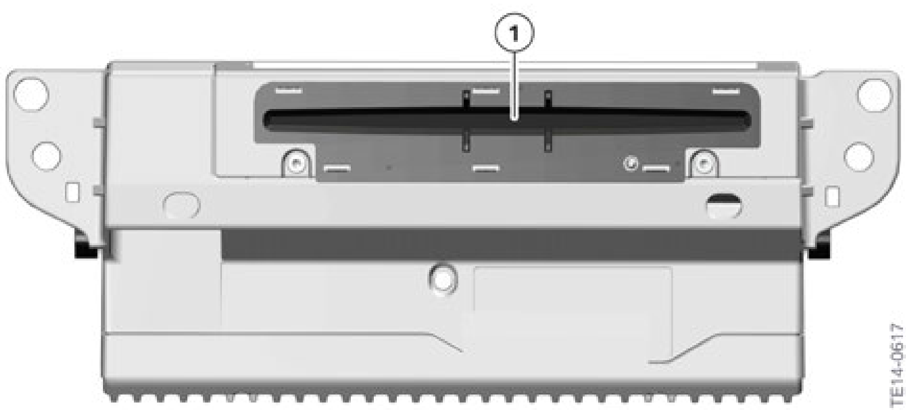
The Headunit High 2 is the model that succeeds the highly successful Headunit High. The F23 is the first vehicle in which the new Headunit High 2 is installed.

Operation and the screen interface in the F23 are adopted from the established Headunit High.

The Headunit High 2 is characterized by the following features:

- 1.5” DIN device
- Hard disk with 200 GB storage capacity (adopted from Headunit High)
- New telematic control unit Telematic Communication Box 2 (TCB2)

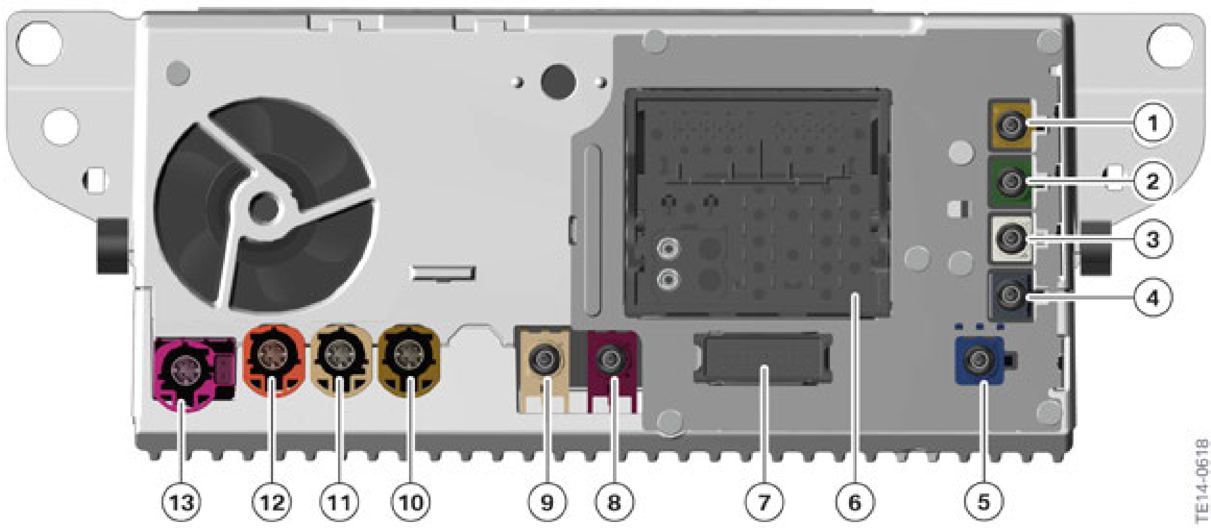
**Front view**




Front view, Headunit High 2

| Index | Explanation |
|---|---|
| 1 | DVD drive |

#### Rear view




Rear view, Headunit High 2

| Index | Explanation |
|---|---|
| 1 | DAB Band III antenna (Not US) |
| 2 | DAB L Band antenna (Not US) |
| 3 | FM2 antenna |
| 4 | AM/FM1 antenna |
| 5 | GPS antenna (only used on vehicles without the optional equipment BMW assist eCall (Option 6AC) or BMW TeleServices (Option 6AE)) |
| 6 | Connection, main connector |
| 7 | Ethernet connection; connection to the Telematic Communication Box 2 (TCB2) |
| 8 | Connection for WLAN antenna |
| 9 | Connection for Bluetooth antenna |
| 10 | USB2 connection; customer's smartphone via the telephone base plate |
| 11 | USB1 connection; customer access to USB audio interface |
| 12 | APIX connection to instrument cluster (not F23) |
| 13 | APIX connection to central information display (CID) |

## 6.2. Updating map data

### 6.2.1. Updating by the workshop

Updating the map data remains unchanged for BMW Service. BMW Service can load the map data either via the USB interface or via the BMW diagnosis system.

### 6.2.2. Updating by the customer

The customer can use the following options to update his map data:

- Data transfer via the internal vehicle interface
- Automatic updating (only in conjunction with the Headunit High 2).

**USB interface**


Map data updating via the USB interface

| Index | Explanation |
|---|---|
| 1 | Customer buys the map data on a USB stick at the BMW dealership |
| 2 | USB stick with the current map data |
| 3 | Customer loads the map data by USB stick and enabling code via the USB interface to the vehicle |

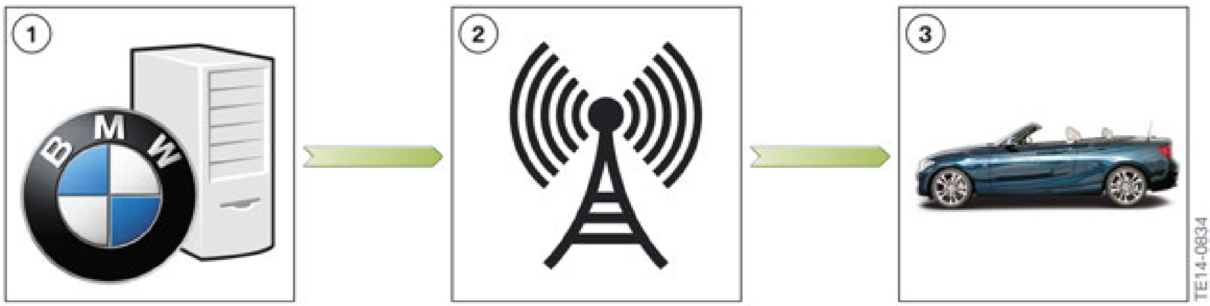
**Automatic updating**

An additional option for updating the map data is available to the customer on vehicles with the Headunit High 2 and the optional equipment BMW TeleServices (Option 6AE). This is done automatically via an online connection.

The complete data is not updated (approx. 30 GB); instead, only the current changes for the home region (max. 350 MB) are updated.

Updating is done automatically, i.e. the customer decides or ops not have to start the updating manually.

The map data is updated regularly and several times a year when updated map data (not the full version) is available. The progress of the update can be viewed in the central information display (CID) under Options; installation is performed conveniently and fully automatically.




Automatic map data updating

| Index | Explanation |
|---|---|
| 1 | BMW server with the current map data |
| 2 | Updated map data (not a full update) are transferred online to the customer's vehicle |
| 3 | Customer's vehicle with the optional equipment BMW TeleServices (Option 6AE) |

Only the map data of the respective home region (e.g. USA) are updated.

The minimum requirement for automatic updating is terminal 15 active. If the updating is aborted (e.g. no data connection or end of journey) the updating is continued as soon as terminal 15 is active and the vehicle has re-established a data connection.

**Automatic updating will be available from Fall 2014 in the following markets:**

- Germany
- Great Britain/Ireland
- Italy
- France
- Netherlands
- Spain
- US
- Canada
- China

## 6.3. Telematic control units

The F23 has different telematic control units, depending on the equipment installed. These differ in where they are installed and how they function.

### 6.3.1. Variants

The table below shows which telematic control units are installed for the respective headunits:

| Headunit/Radio | Telematic control unit |
|---|---|
| BMW Professional radio (standard equipment) | Telematic Communication Box (TCB) |
| Professional navigation system (Option 609) | Telematic Communication Box 2 (TCB2) |

### 6.3.2. Functions

| Function | Telematic Communication Box (TCB) | Telematic Communication Box 2 TCB2 |
|---|---|---|
| BMW Assist eCall (Option 6AC) | X | X |
| BMW TeleServices (Option 6AE) | X | X |
| Remote Services (Option 6AP) | X | X |
| BMW Online and BMW Apps (Option 6AK) | X | X |
| Data transfer rate | 3G | 3G or 4G* |

* 4G (LTE) only in conjunction with the optional equipment Convenience telephone with extended smartphone connectivity in the US version.

### 6.3.3. Telematic Communication Box (TCB)

In conjunction with a Headunit Basis the Telematic Communication Box (TCB) familiar from other BMW models is installed.

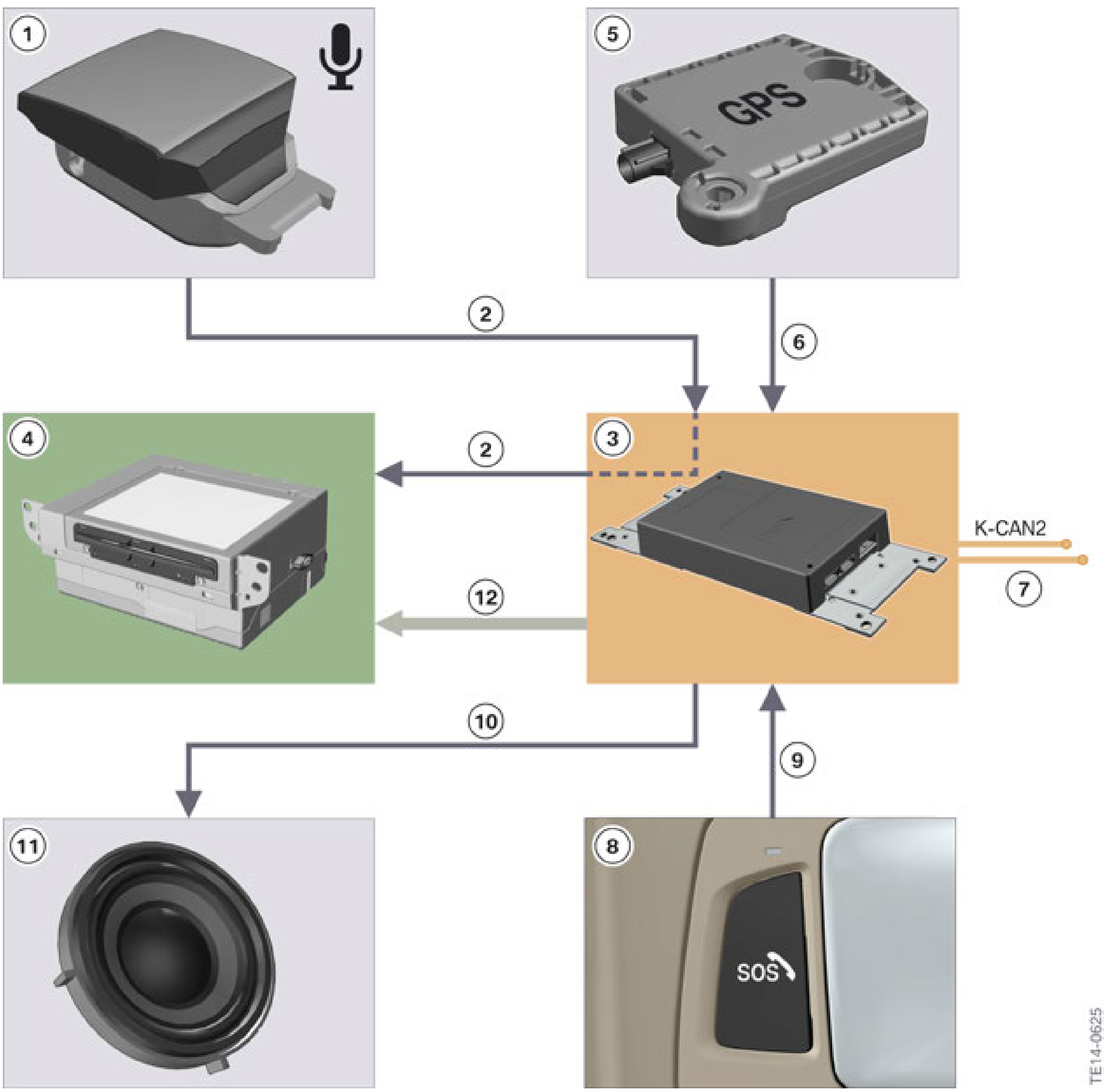
Further information on the Telematic Communication Box (TCB) can be found in the Training Reference Material “Headunit High” from the year 2012.

### 6.3.4. Telematic Communication Box 2 (TCB2)

#### Function

The Telematic Communication Box 2 (TCB2) is identified by the following features:

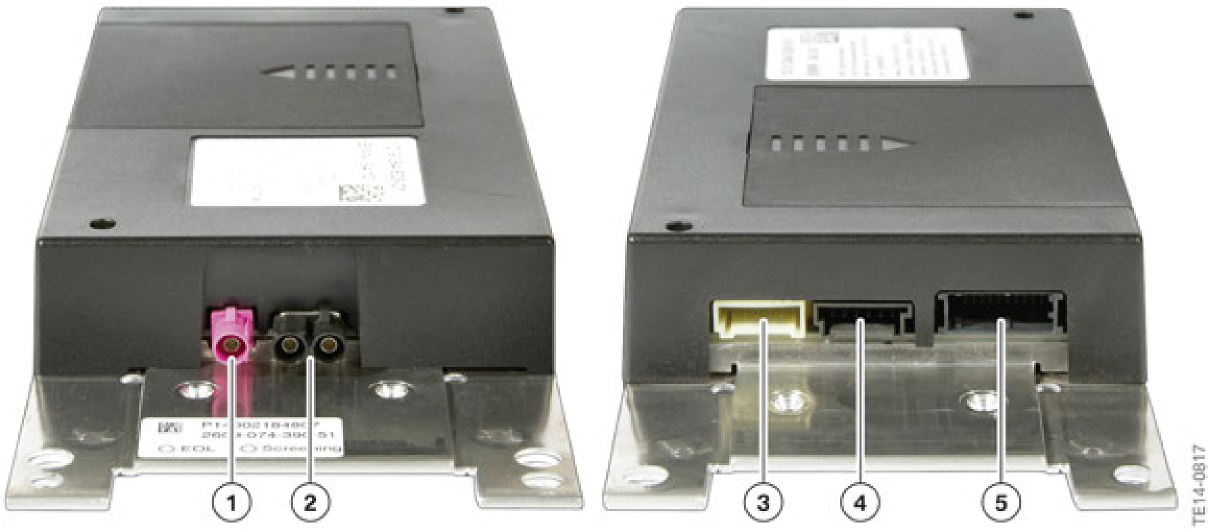
- Ethernet connection to the headunit when a Headunit High 2 (HU-H2) and a Telematic Communication Box 2 (TCB2) are installed at the same time, the GPS data are received by the TCB2 and made available by the Ethernet connection to the HU-H2
- The driver's side microphone is connected to the TCB2; the microphone signal is provided via the TCB2 for the headunit.




F23 functionTCB2

| Index | Explanation |
|---|---|
| 1 | Steering column microphone |
| 2 | Microphone signal |
| 3 | Telematic Communication Box 2 (TCB2) |
| 4 | Headunit High 2 |
| 5 | GPS antenna |
| 6 | GPS signal |
| 7 | K-CAN2 |
| 8 | Emergency call button |
| 9 | Emergency call button connection |
| 10 | SOS (emergency) loudspeaker connection |
| 11 | SOS (emergency) loudspeaker |
| 12 | Ethernet connection |

#### Connections



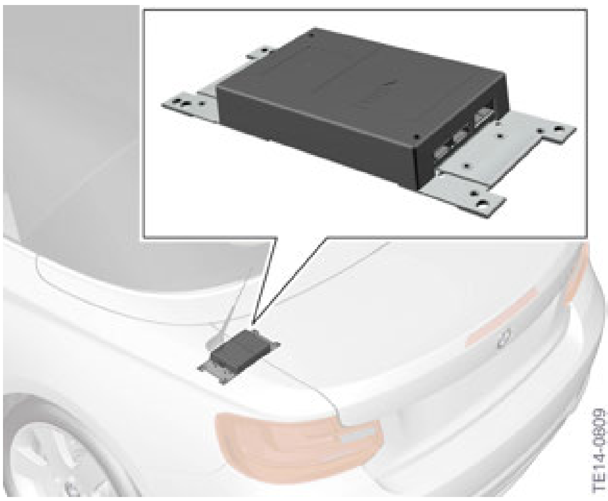

F23 connections TCB2

| Index | Explanation |
|---|---|
| 1 | GPS antenna |
| 2 | • Telematics antenna (TEL2)<br>• Emergency GSM antenna |
| 3 | • SOS (emergency) loudspeaker<br>• Terminal 30F<br>• Ground |
| 4 | • Emergency call button<br>• LED, emergency call button<br>• Crash signal |
| 5 | • OABR Ethernet<br>• K-CAN2<br>• Driver's microphone |

⚠

The plug-in contacts of the TCB2 must be carefully unlocked and disconnected, otherwise there is a risk of damage.

**Installation location**




F23 Installation locationTCB2

The TCB2 is installed in the luggage compartment.

### 6.3.5. Ethernet

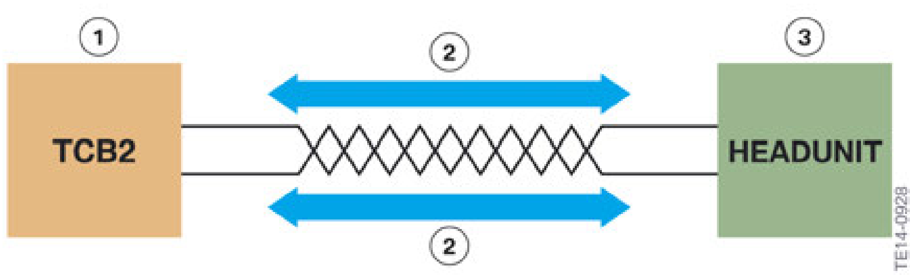
Unshielded Twisted Single Pair is an Ethernet variant which was developed by the OPEN Alliance Broad-Reach. This refers to two single-wire, twisted and unshielded lines for data transfer.

In the F23 the TCB2 is connected to the headunit via OABR Ethernet.

The data rate for an OABR Ethernet connection is max. 2 x100 Mbit/s.

The Ethernet connection from the OBD2 interface to the Body Domain Controller is still via 5 lines.

Because only 2 lines and not 5 lines are used for an OABR Ethernet connection, the weight of the lines can thus be reduced.




OABR Ethernet F23

| Index | Explanation |
|---|---|
| 1 | Telematic Communication Box 2 (TCB2) |
| 2 | Transmit/Receive max. 2 x 100 Mbit/s |
| 3 | Headunit |

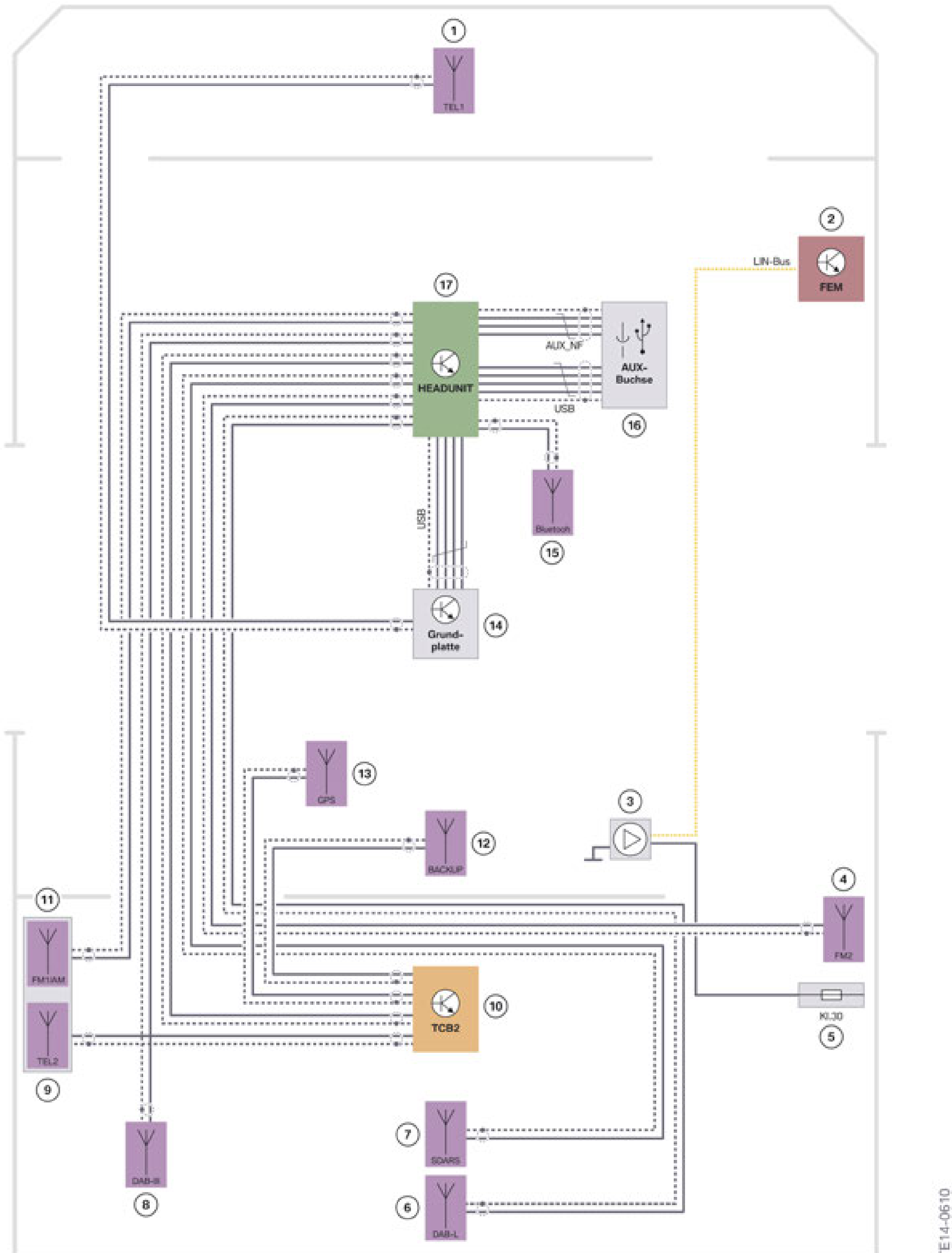
## 6.4. Aerial systems

The F23 has different antenna systems, depending on the national-market version and optional equipment used:

### 6.4.1. Overview

| Aerial | System | Installation location |
|---|---|---|
| FM antenna | Radio | • FM1 rod antenna<br>• FM2 side panel antenna, rear right |
| AM antenna | Radio | Rod antenna |
| DAB antenna (Not US) | Radio | • L band on the tailgate<br>• Band III bumper, rear left |
| SDARS antenna | Radio | Trunk lid |
| GPS antenna | Navigation system | Under the soft top compartment lid |
| Remote control services antenna | Front Electronic Module (remote control services) | On the rollover protection system |
| Telephone antenna | Telephone | Bumper, front |
| Telephone antenna, telematics services | Telematics services | Integrated in the rod antenna |
| Bluetooth antenna | Telephone | Wiring harness |
| Emergency GSM antenna | Telematics services | Under the seat bench |
| VICS antenna | Japan navigation system | Inside mirror |

### 6.4.2. System wiring diagram for antenna systems



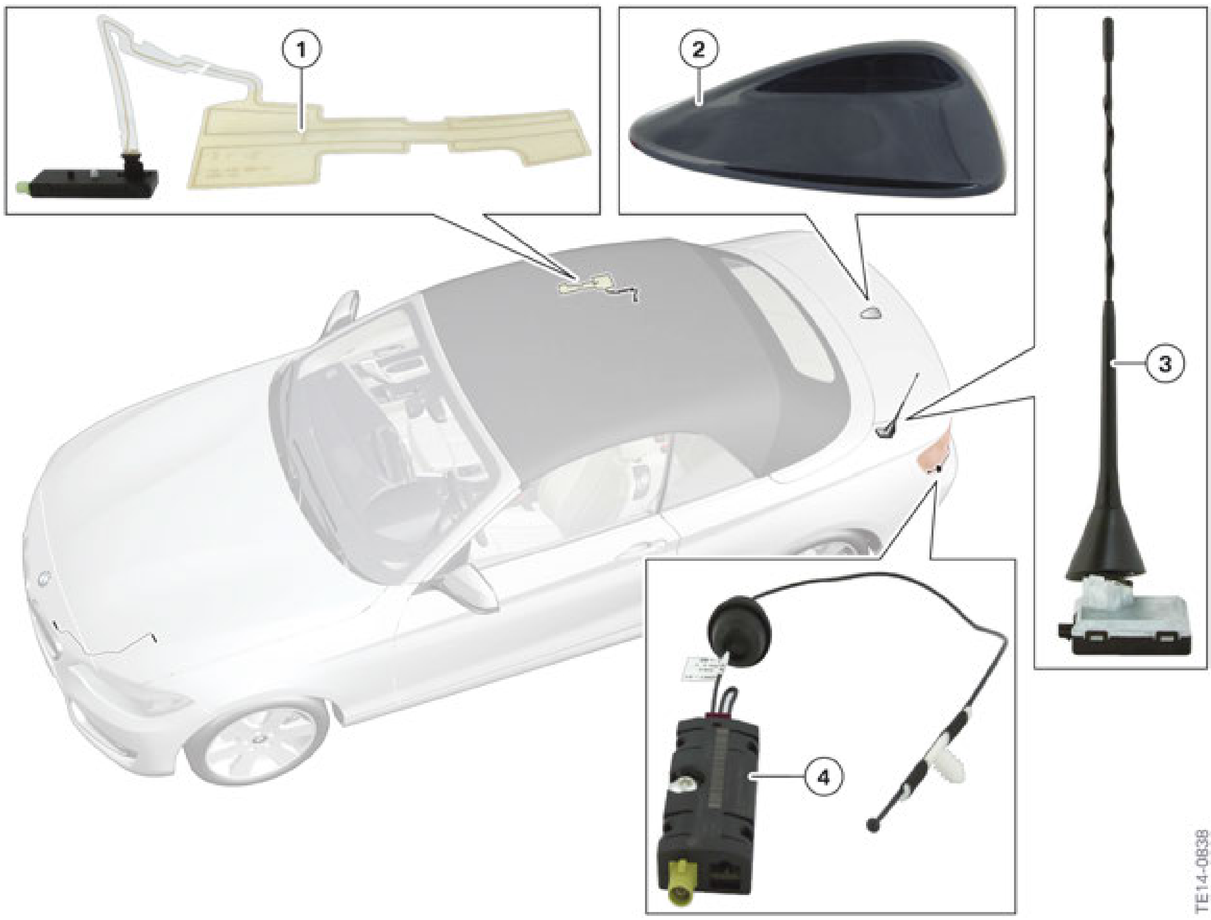

F23 System wiring diagram for antenna systems

# F23 Complete Vehicle
## 6. Info and Communication Systems

| Index | Explanation |
| --- | --- |
| 1 | Telephone antenna for base plate (TEL1) |
| 2 | Front Electronic Module (FEM) |
| 3 | Remote control receiver |
| 4 | FM2 antenna |
| 5 | Power distribution box, luggage compartment |
| 6 | DAB L Band antenna (Not US) |
| 7 | SDARS antenna |
| 8 | DAB Band III antenna (Not US) |
| 9 | Telephone antenna, telematics services (TEL2) |
| 10 | Telematic Communication Box 2 (TCB2) |
| 11 | AM/FM1 antenna |
| 12 | Emergency GSM antenna |
| 13 | GPS antenna |
| 14 | Base plate (TEL1) |
| 15 | Bluetooth antenna |
| 16 | USB audio interface |
| 17 | Headunit |

### 6.4.3. System components

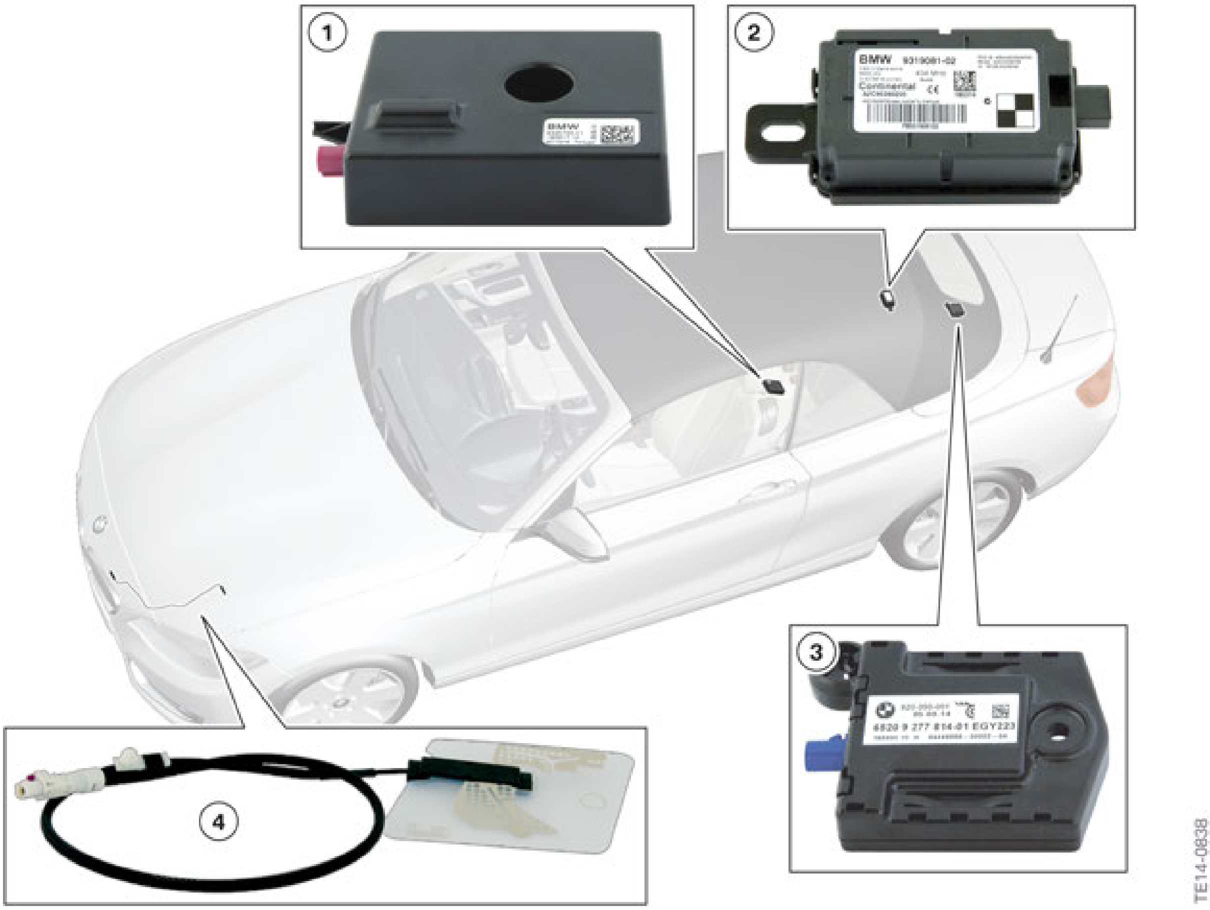
The following graphic shows the installation locations of the individual antenna system components in the F23.




F23 antenna systems

| Index | Explanation |
|---|---|
| 1 | FM2 antenna (side panel antenna) |
| 2 | SDARS antenna |
| 3 | FM1/AM antenna/telephone antenna, telematics services (TEL2) |
| 4 | DAB Band III antenna (Not US) |

# F23 Complete Vehicle
## 6. Info and Communication Systems




F23 antenna systems

| Index | Explanation |
|---|---|
| 1 | Emergency GSM antenna |
| 2 | Remote control receiver |
| 3 | GPS antenna |
| 4 | Telephone antenna (TEL1) |

## 6.5. Speaker systems

The F23 has two various speaker systems:

- Hi-fi speaker system (Option 676)
- Harman Kardon® hi-fi speaker system (Option 674)

### 6.5.1. Performance

| System | Overall power |
|---|---|
| Hi-fi speaker system (Option 676) | 205 W |
| Harman Kardon® hi-fi speaker system (Option 674) | 360 W |

### 6.5.2. Speaker

The table below shows the number of different speakers and the installation and connection of an amplifier (AMP):

| System | Amplifier (AMP) | Tweeter | Mid-range speaker | Woofer |
|---|---|---|---|---|
| Hi-fi system (Option 676) | Yes (not with diagnostic capability) | 2 | 5 | 2 |
| Harman Kardon® hi-fi system (Option 674) | Yes (K-CAN2) | 5 | 5 | 2 |

### 6.5.3. Active Sound Design (ASD)

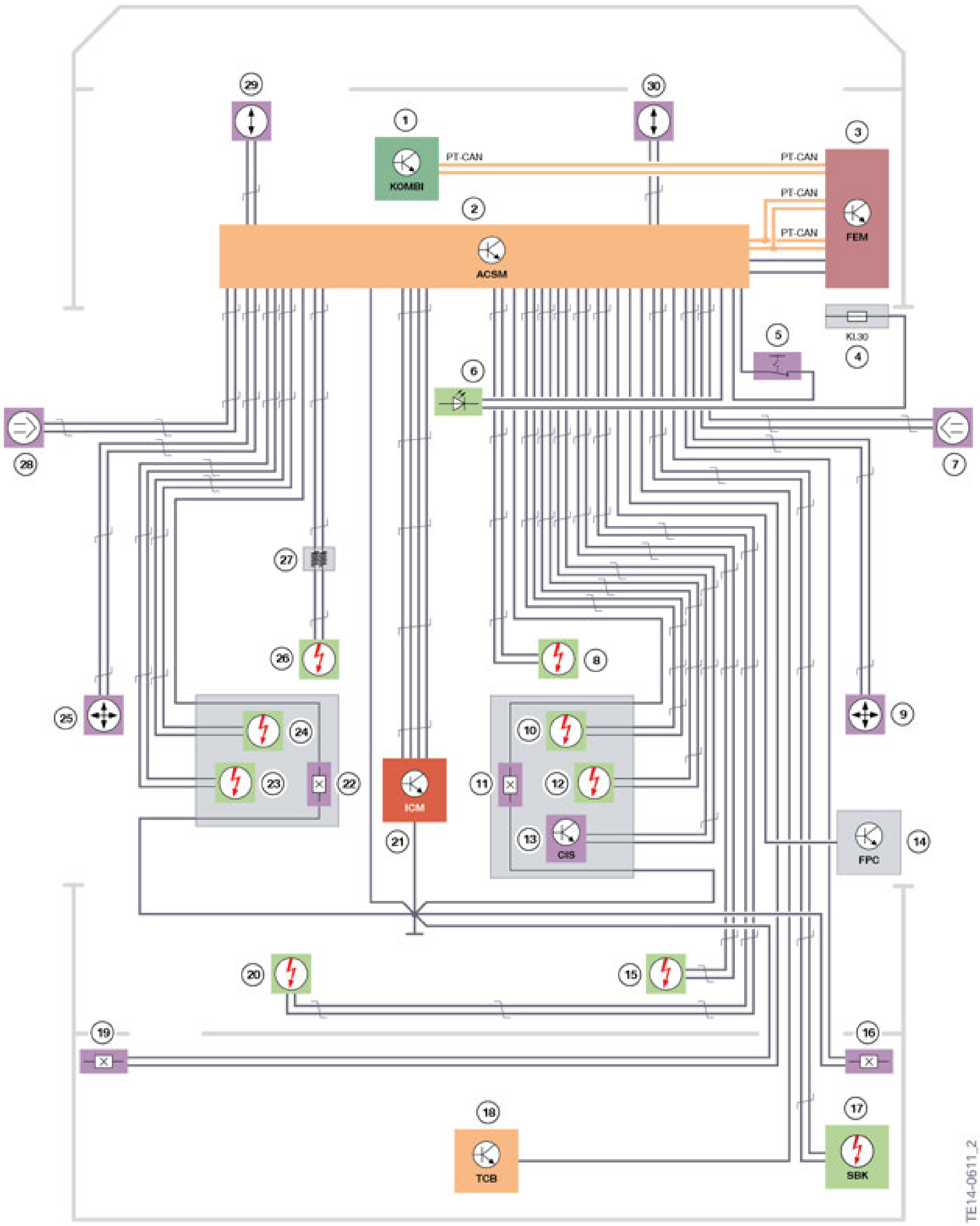
With the N55B30O0 engine the BMW M235i Convertible has an Active Sound Design (ASD), which is integrated in the audio system of the vehicle. The engine noise is enhanced due to the excellent noise insulation in the passenger compartment. The Active Sound Design conducts an engine sound generated in the passenger compartment, thus specifically adding to the driving pleasure.

Further information on ASD can be found in the Technical Training Reference Manual “F10 M Complete Vehicle”.

# F23 Complete Vehicle
## 7. Passive Safety

Because the body rigidity on the F23 is higher as on other BMW Convertibles, additional airbag sensors are installed in the doors. These additional pressure sensors increase the safety of the F23 in the event of an accident. The US version uses two up front sensors, duel stage front airbags with vent, front knee air bags and a CIS mat for passenger front seat. Please refer to the proper wiring diagram for more details.

### 7.1. System wiring diagram




F23 system wiring diagram for “Passive Safety” (Incomplete for US market)

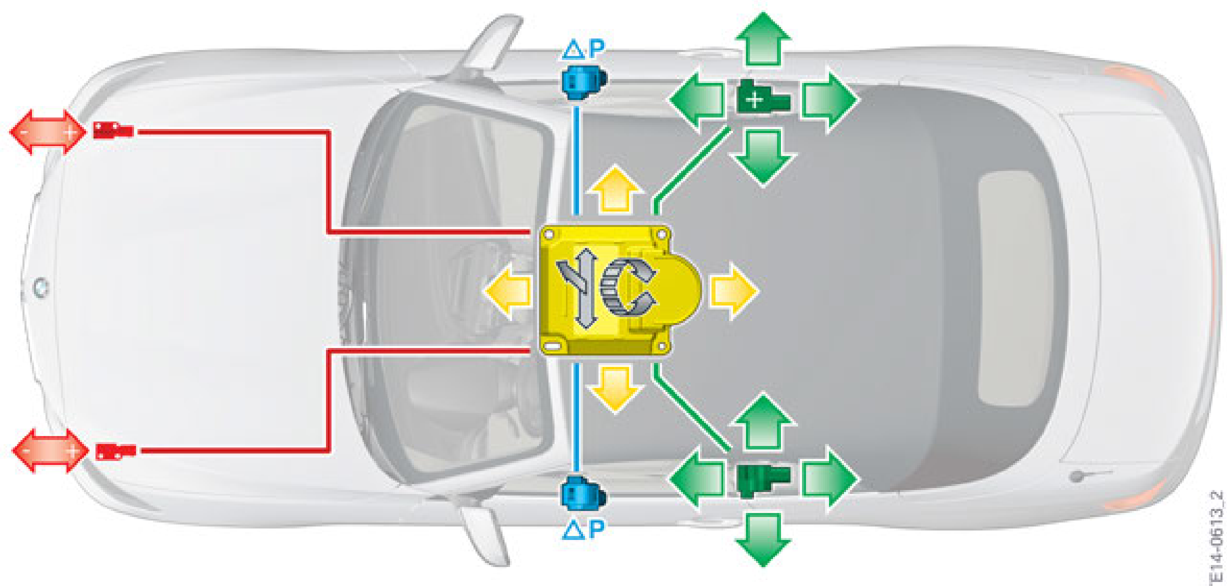
# F23 Complete Vehicle
## 7. Passive Safety

| Index | Explanation |
|---|---|
| 1 | Instrument cluster |
| 2 | Crash Safety Module (ACSM) |
| 3 | Front Electronic Module (FEM) |
| 4 | Power distribution box, front passenger footwell |
| 5 | Switch, front passenger airbag deactivation |
| 6 | Indicator light, front passenger airbag deactivation |
| 7 | Airbag sensor, door, right |
| 8 | Front passenger airbag |
| 9 | Acceleration sensor, side panel, right |
| 10 | Side airbag, front passenger |
| 11 | Seat belt buckle contact, front passenger |
| 12 | Belt tensioner, front passenger |
| 13 | Seat occupancy detection European version (SBE); US version (CIS); |
| 14 | Fuel pump control (FPC) |
| 15 | Electromagnetic actuator, right (rollover protection system) |
| 16 | Seat belt buckle contact, rear right |
| 17 | Safety battery terminal (SBK) |
| 18 | Telematic Communication Box (TCB) |
| 19 | Seat belt buckle contact, rear left |
| 20 | Electromagnetic actuator, left (rollover protection system) |
| 21 | Integrated Chassis Management (ICM) |
| 22 | Seat belt buckle contact, driver |
| 23 | Belt tensioner, driver |
| 24 | Side airbag, driver's side |
| 25 | Acceleration sensor, side panel, left |
| 26 | Driver's airbag |
| 27 | Clock spring |
| 28 | Airbag sensor, door, left |
| 29 | Up front sensor, left |
| 30 | Up front sensor, right |

## 7.2. Sensors

The US version of the F23 is equipped with the following sensors:

- one lateral and one longitudinal acceleration sensor in the rear side panels (green)
- one airbag sensor to monitor the pressure in each of the front doors (blue)
- one lateral and one longitudinal acceleration sensor in the ICM (yellow)
- one roll rate sensor in the ICM (yellow)
- two bumper mounted up front sensors (red) hard wired connection to the ACSM control unit




US version of the F23 sensors

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